



ASUS ZA550KL service manual

Technical documents

file name : **ASUS ZA550KL** service manual

version : V1.0



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CMU200 or 8960



DC power supply



Oscilloscope



Hot air gun



Electric iron



Spectrum Analyzer



electric screwdriver



UP-1024 high speed programmer



Removal tool



Removal tool



multimeter



Anti-static gloves



Lead-free solder wire



Anti-static cloth



alcohol bottle



brush



Anti-static brush



Anti-static fixture

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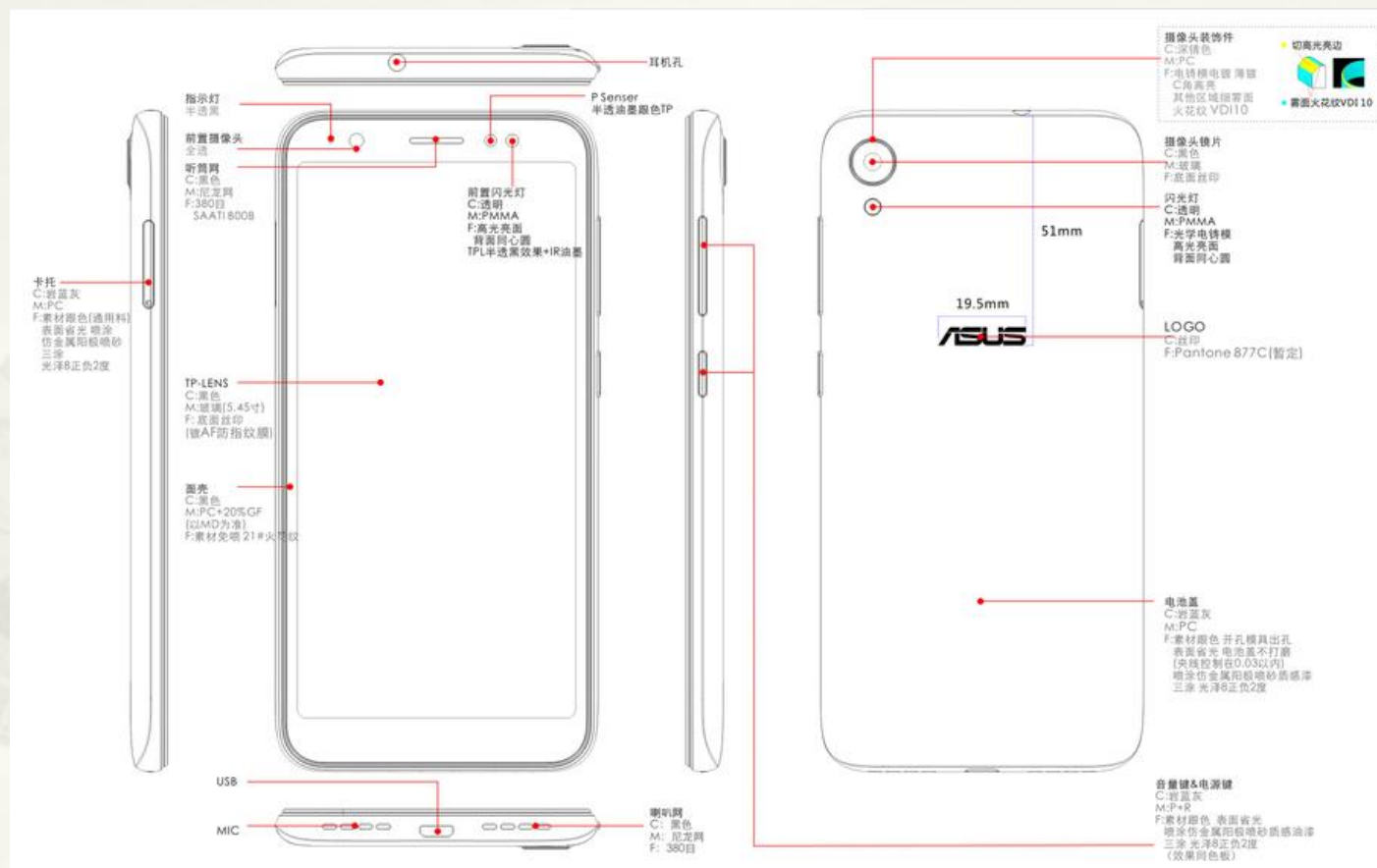
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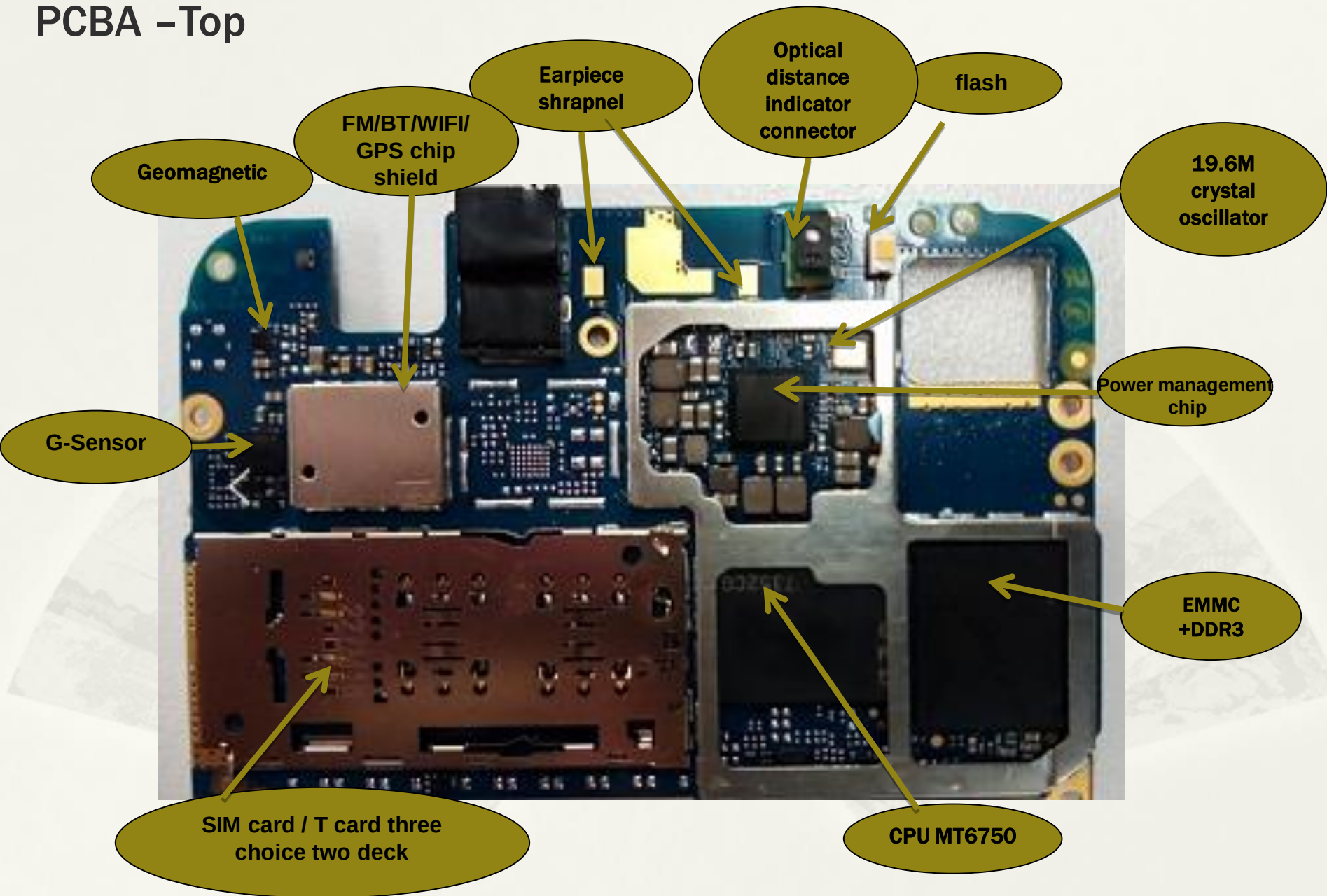
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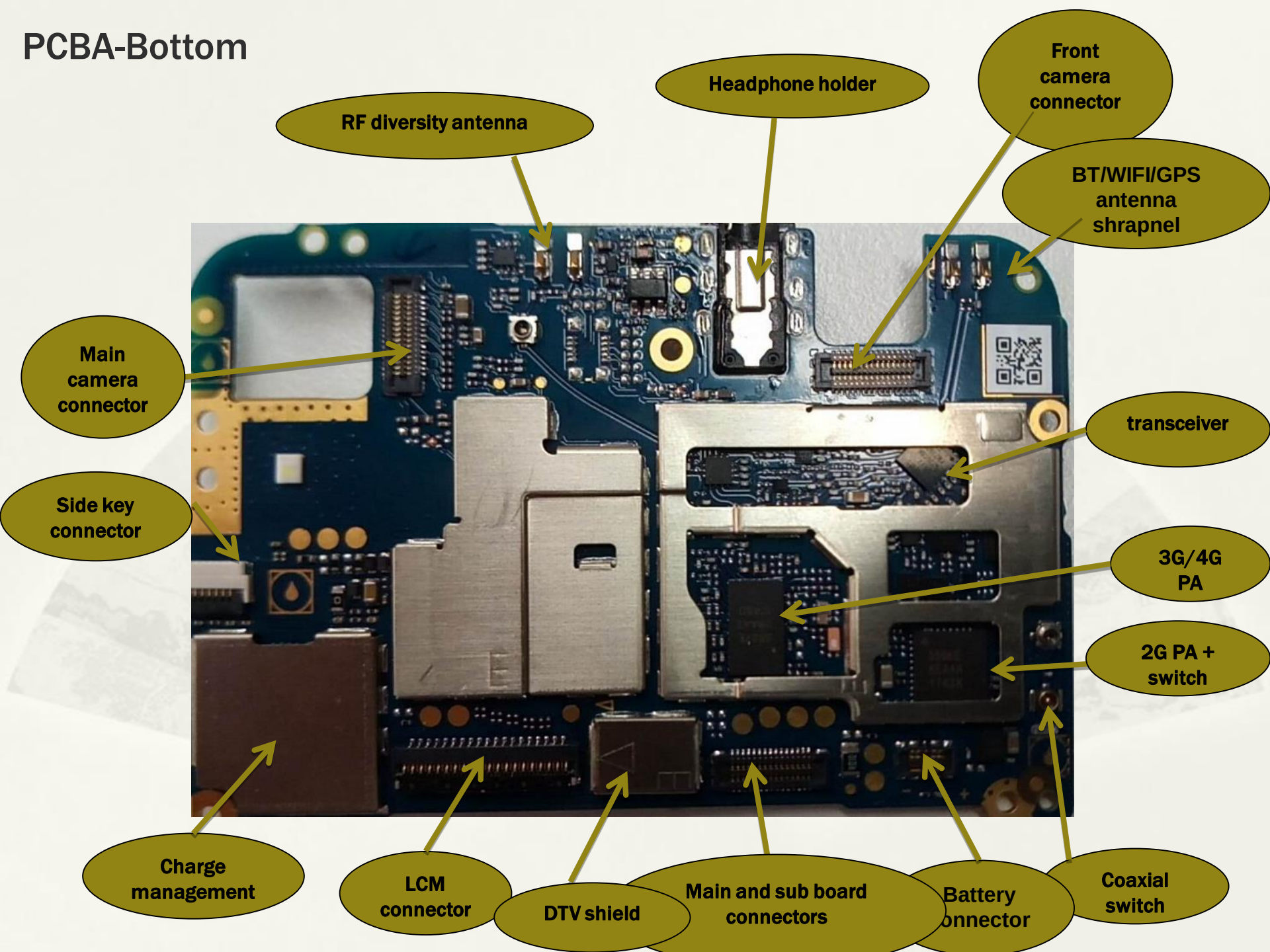
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PCBA -Top



PCBA-Bottom



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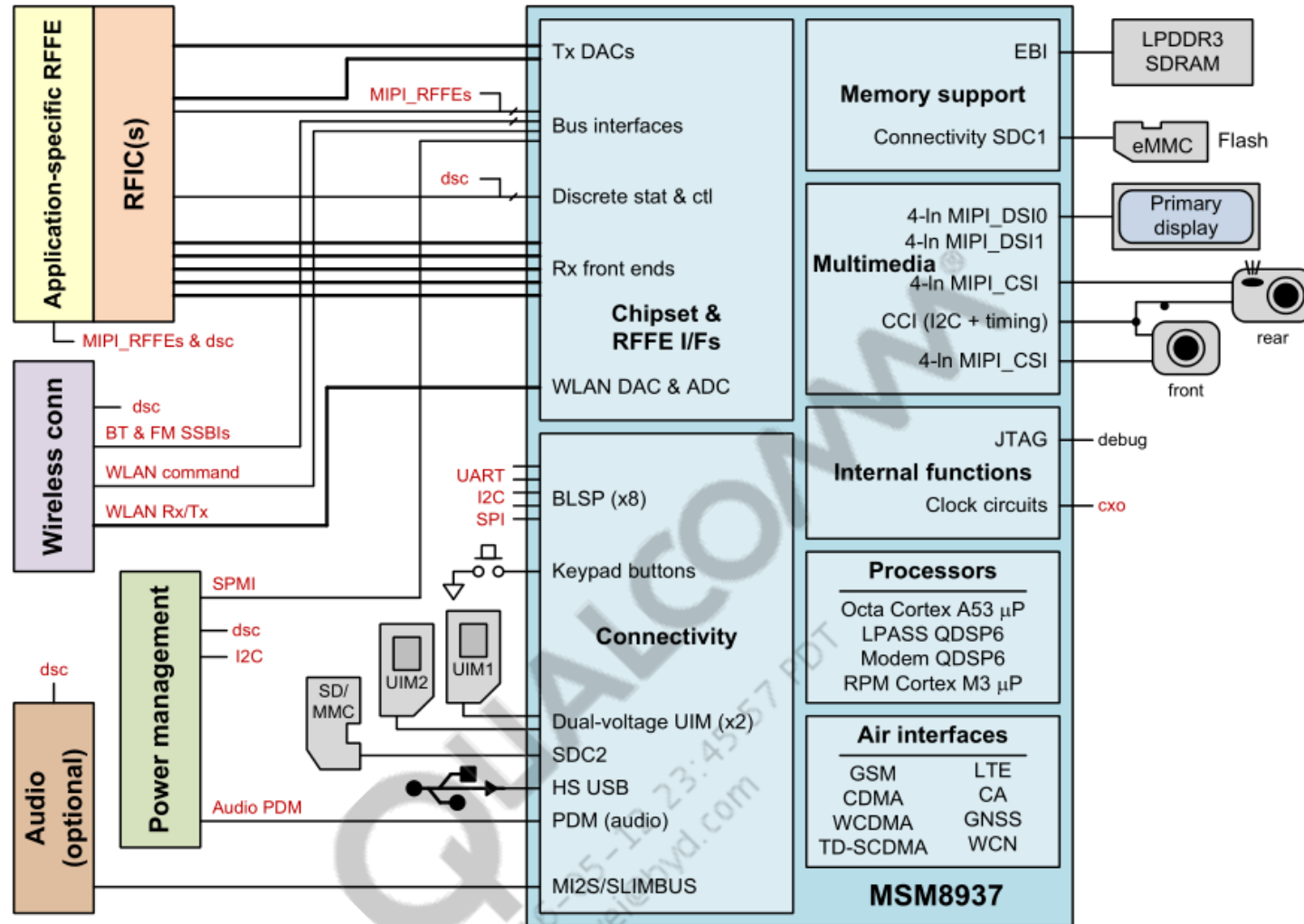
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Button switch timing

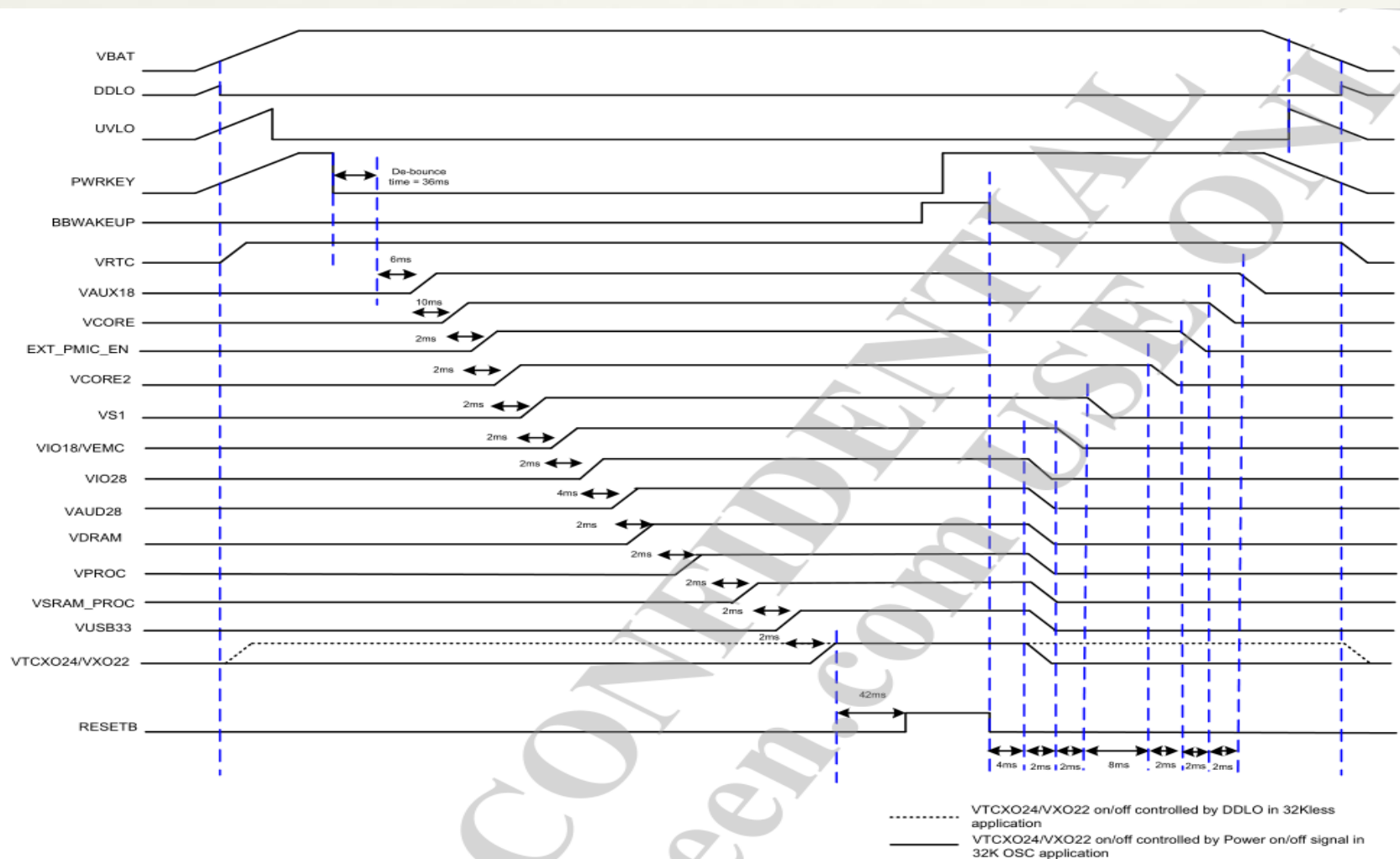


Figure 3-2. Power-on/off control sequence with XTAL by pressing PWRKEY

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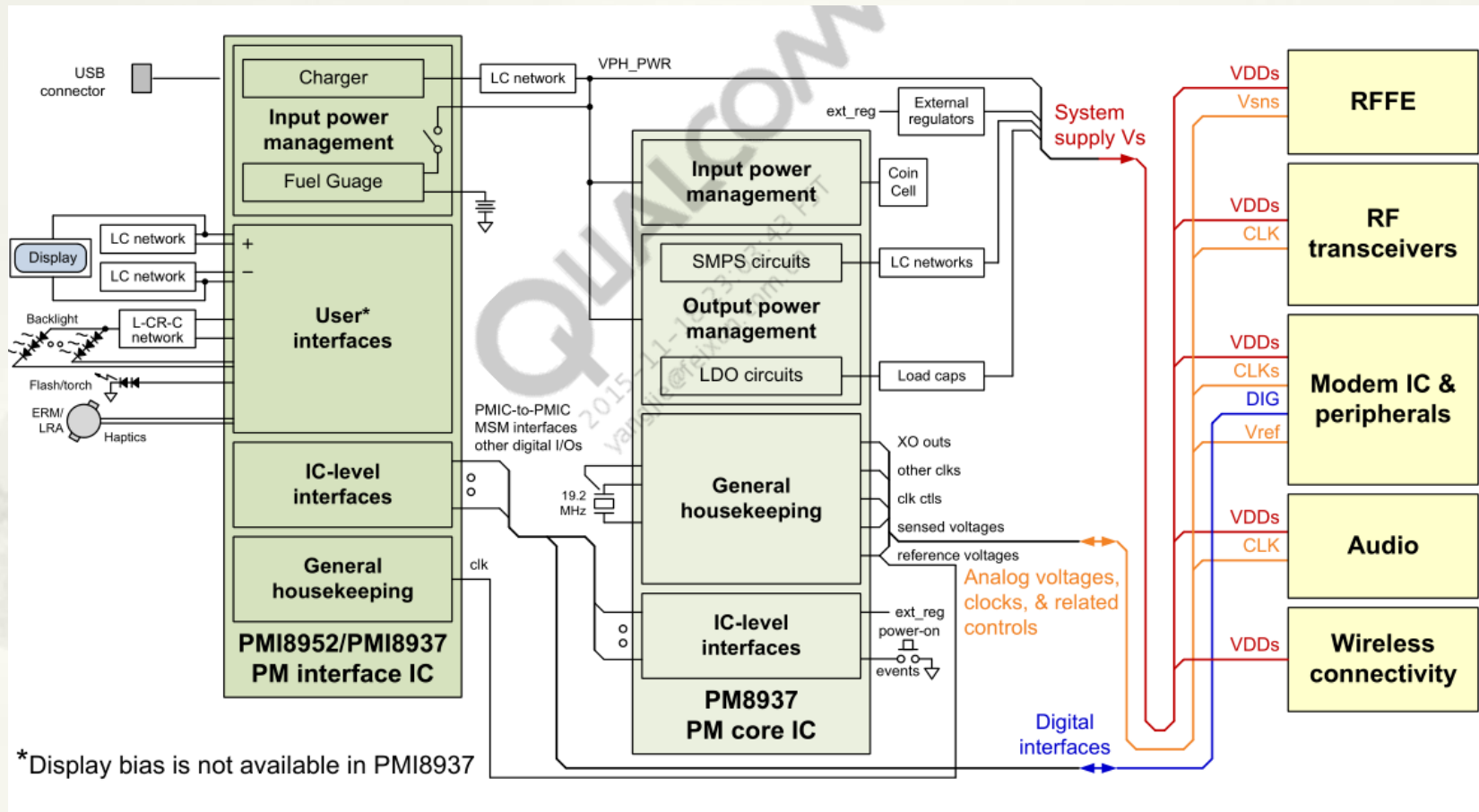
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I2C address

Acceleration sensor	\	INVESENSE	ICM-20608D	I2C4	Write:0xD0, Read:0xD1
Geomagnetic sensor	\	VTC	AF8133J	I2C4	Write:0x3C, Read:0x3D
		MEMSIC	MMC3680KJ	I2C4	
Gyro sensor	\	ST	LSM6DS33	I2C4	Write:0xD4, Read:0xD5
		INVESENSE	ICM-20608D	I2C4	Write:0xD0, Read:0xD1
Three-in-one sensor	\	LITEON	LTR-578ALS-039	I2C4	Write:0xA6 \Read:0xA7
SAR sensor	\				
front camera	Rear photography 8M	Omnivision	OV8856	I2C1	0X20/0X21
		Hynix	HI846		0X40/0X41
	Front photography 8M	Hynix	HI846	I2C1	0X46/0X47
		Omnivision	OV8856		0X6C/0X6D
	Front photography 5M	Hynix	HI1332	I2C1	0X40/0X41
		Omnivision	OV13858		0X6C/0X6D
Rear camera Driver ID	Rear photography 8M	Giantec	CN3937	I2C0	0X18/0X19
		Giantec	GT9762		0X18/0X19
	Rear photography 8M	Giantec	GT24C64A	I2C1	0XA0/0XA1
		Giantec	GT24C64A		0XA0/0XA1
	Front photography 5M	Giantec	GT24C64AH	I2C1	0XA8/0XA9
		Giantec	GT24C64AH		0XA8/0XA9
LCD				GPI010 , 11	
				GPI010 , 11	
				GPI010 , 11	
touch screen				GPI010 , 11	

Power management



Buck Power management

Function	Circuit type	Default V (V) ¹	Specified range (V) ² (MSM8937)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8937)
S1	ULT-SMPS	1.225	0.900–1.350	0.375–1.5625	2000 ³	N	MSM modem
S2	ULT-SMPS	1.225	0.550–1.350	0.375–1.5625	3000	Y	MSM core and graphics
S3	HF-SMPS	1.288	1.200–1.4125	0.375–1.5625	2700	Y	Low-voltage LDOs (1, 2, 3, 19, and 23)
S4	ULT-SMPS	2.050	1.800–2.050	1.550–3.125	2500	Y	High-voltage LDOs (4, 5, 6, 7, 16, RFCLK, and XO)
S5	FT-SMPS	1.225	1.050–1.350	0.350–1.355	3000	Y	MSM applications processor
S6	FT-SMPS	1.225	1.050–1.350	0.350–1.355	3000	N	MSM applications processor

LDO Power management

L1	NMOS LDO	1.000	1.000	0.375–1.5375	1200	N	RFICs
L2	NMOS LDO	1.200	1.200	0.375–1.5375	1200	Y	LPDDR2/LPDDR3, MIPI CSI, and DSI
L3	NMOS LDO	1.225	0.750–1.350	0.375–1.5375	1200	Y	VDDMX
L4	PMOS LDO	1.800	1.800	1.750–3.3375	450	N	RFICs and GPS eLNA
L5 ⁴	PMOS LDO	1.800	1.800	1.750–3.3375	500	Y	Most digital I/Os, MSM pad groups 3 and 7, LPDDR, and eMMC
L6	PMOS LDO	1.800	1.800	1.750–3.3375	300	N	MSM DSI PLL and OTP, camera, touchscreen, display, and sensors
L7	PMOS LDO	1.800	1.800	1.750–3.3375	150	Y	MSM analog and PLLs, WCN XO, and PM baseband clock driver
L8	PMOS LDO	2.900	2.900	1.750–3.3375	600	Y	eMMC
L9	PMOS LDO	V _{out} = 3.3 V for V _{BAT} > 3.575 V; V _{out} = 3 V for V _{BAT} < 3.575 V	3.000–3.300	1.750–3.3375	600	N	WCN
L10	PMOS LDO	2.8	2.800	1.750–3.3375	150	N	Sensors
L11 ⁵	PMOS LDO	2.950	2.950	1.750–3.3375	800	Y	Micro SD
L12 ⁴	PMOS LDO	2.950	1.800/2.950	1.750–3.3375	150	Y	MSM pad group 2 and SDC2
L13	PMOS LDO	3.075	3.075	1.750–3.3375	50	Y	MSM USB and audio
Function	Circuit type	Default V (V) ¹	Specified range (V) ² (MSM8937)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8937)
L14 ⁵	PMOS LDO	1.800	1.800/2.925/3.050	1.750–3.3375	50	N	MSM pad group 5, dual-voltage UIM1, and NFC
L15 ⁵	PMOS LDO	1.800	1.800/2.925/3.050	1.750–3.3375	50	N	MSM pad group 6 and dual-voltage UIM2
L16	PMOS LDO	1.800	1.800	1.750–3.3375	5	N	PMIC HKADC
L17	PMOS LDO	2.850	2.850	1.750–3.3375	600	N	Camera, display, and touchscreen
L18	PMOS LDO	2.700	2.700	1.750–3.3375	150	N	QTI RF front-end
L19	NMOS LDO	1.350	1.350	0.375–1.5375	1200	N	MSM analog, WCN, and WGR
L20	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC XO circuits
L21	Low-noise LDO	1.74	1.74	1.74–3.3375	5	Y	PMIC RF clock buffers
L22	PMOS LDO	2.800	2.800	1.750–3.3375	300	N	Camera – analog
L23	NMOS LDO	1.3	1.2	0.375–1.5375	300	N	Camera – digital

Charge switch timing

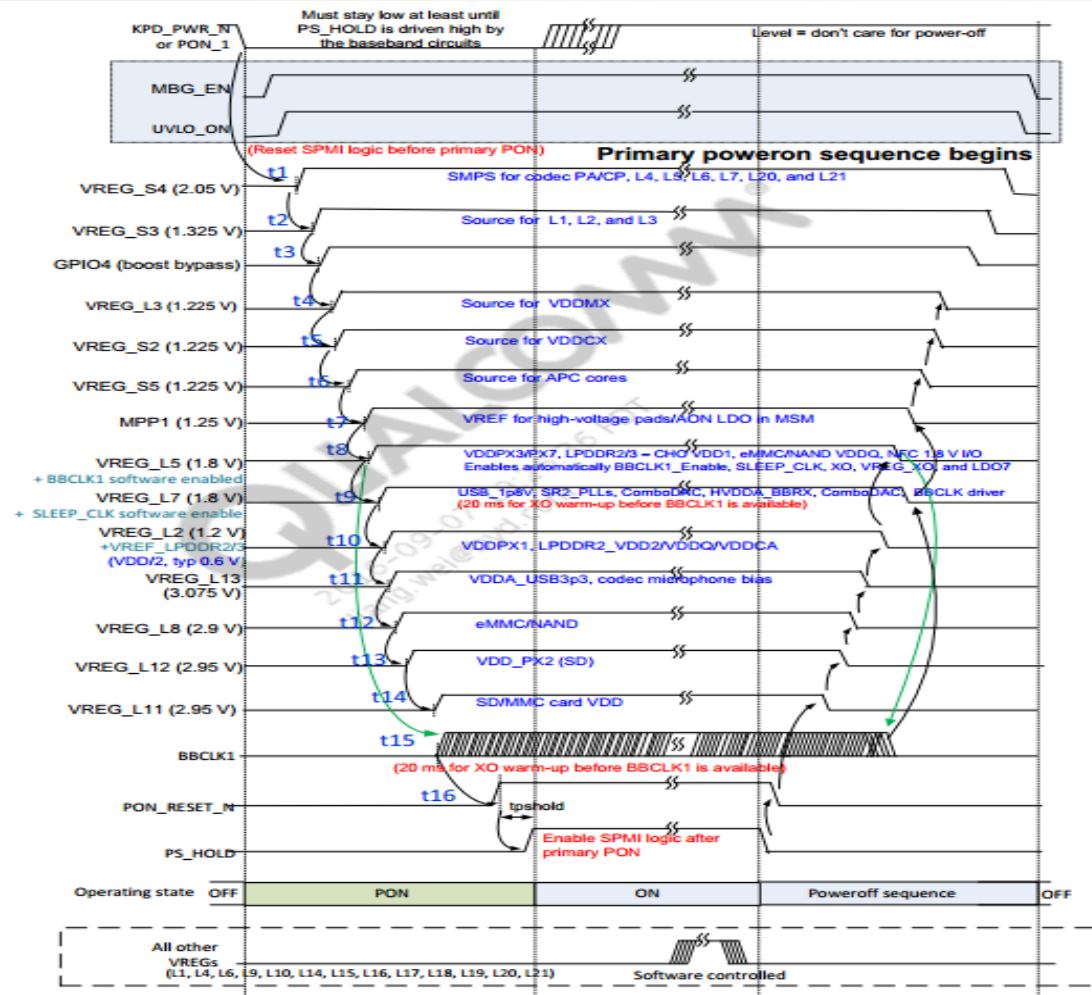


Figure 3-8 PM8937/PM8940 power-on sequence

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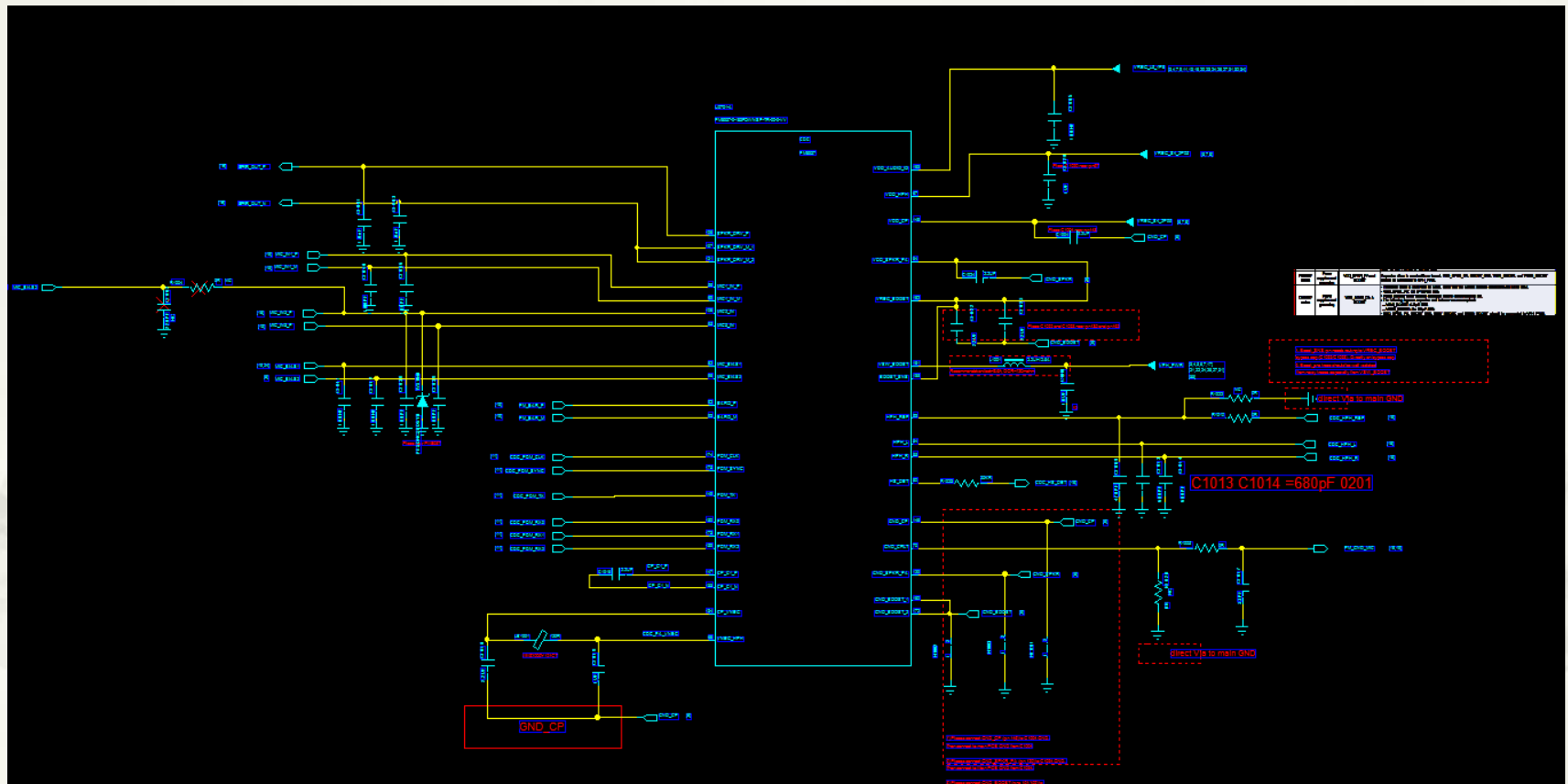
System Block Diagram

I2C address and power management

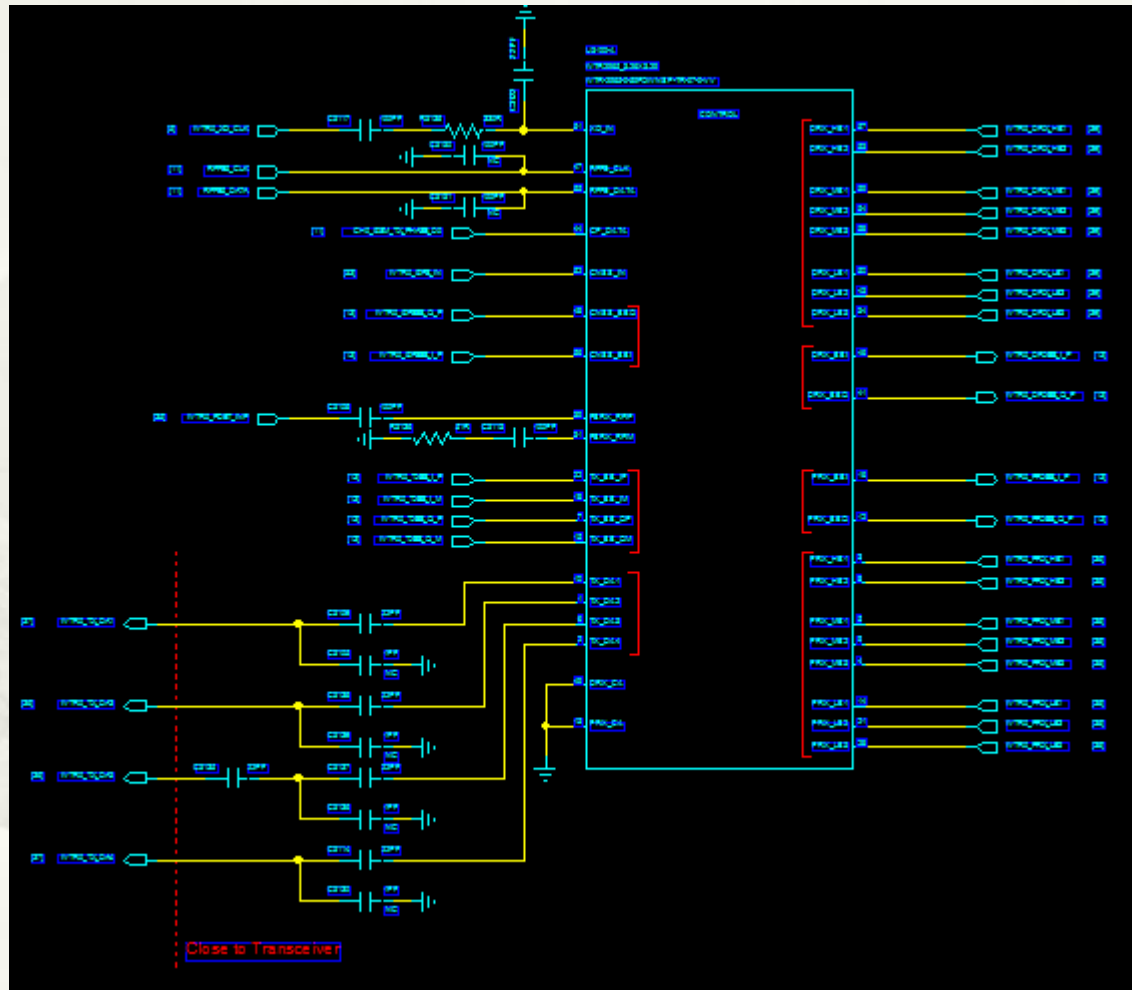
Important devices and key test points

Maintenance method

Power supply : PM8937 U0701

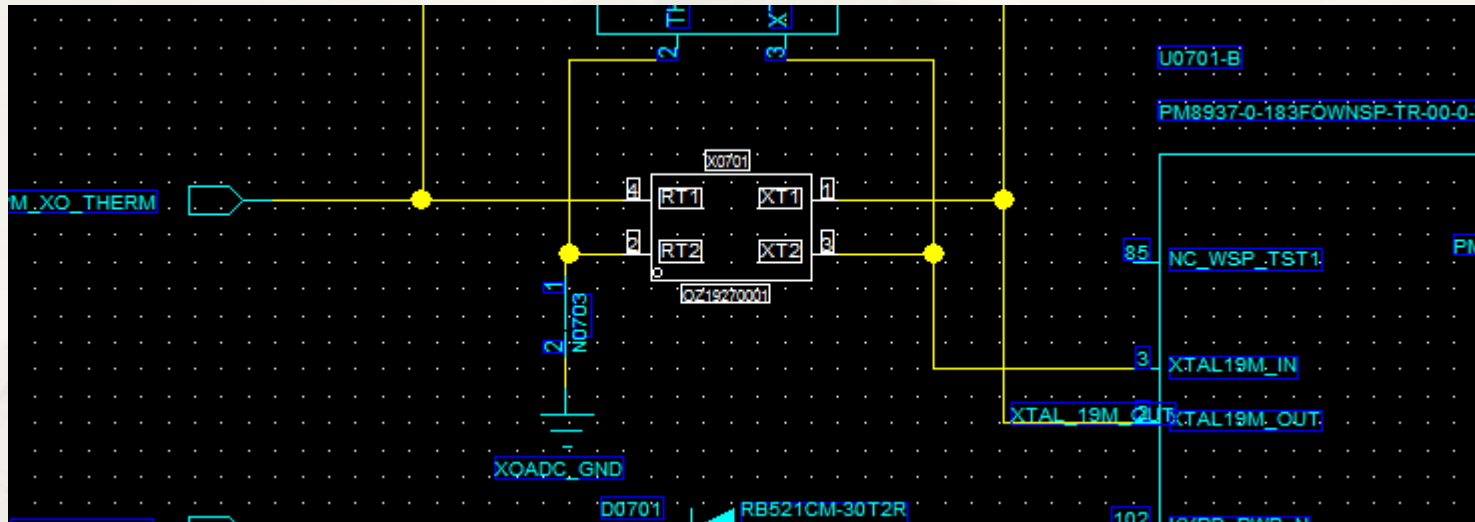


Transceiver WTR2965 U3101

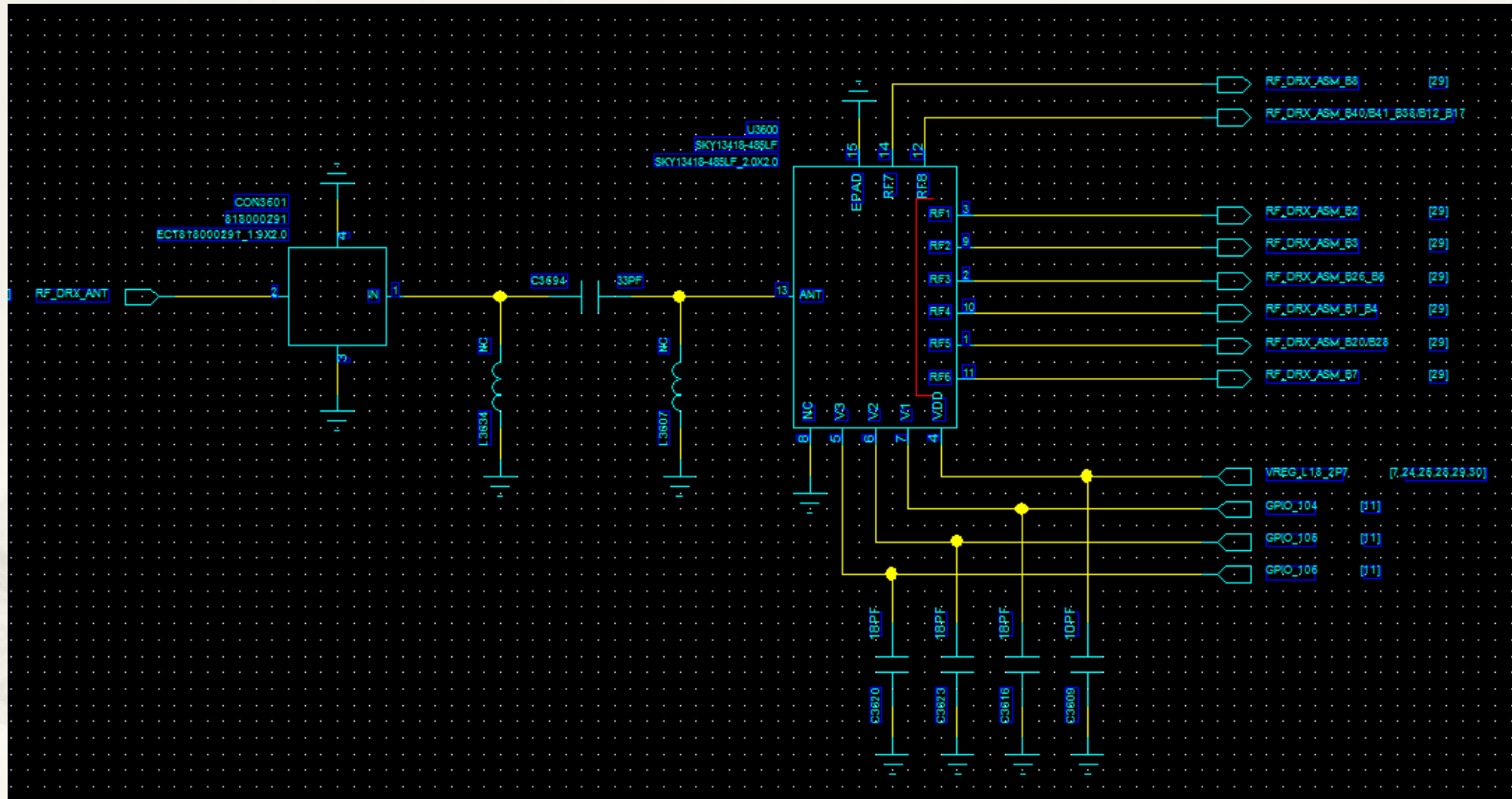


19MHz Crystal section X0701

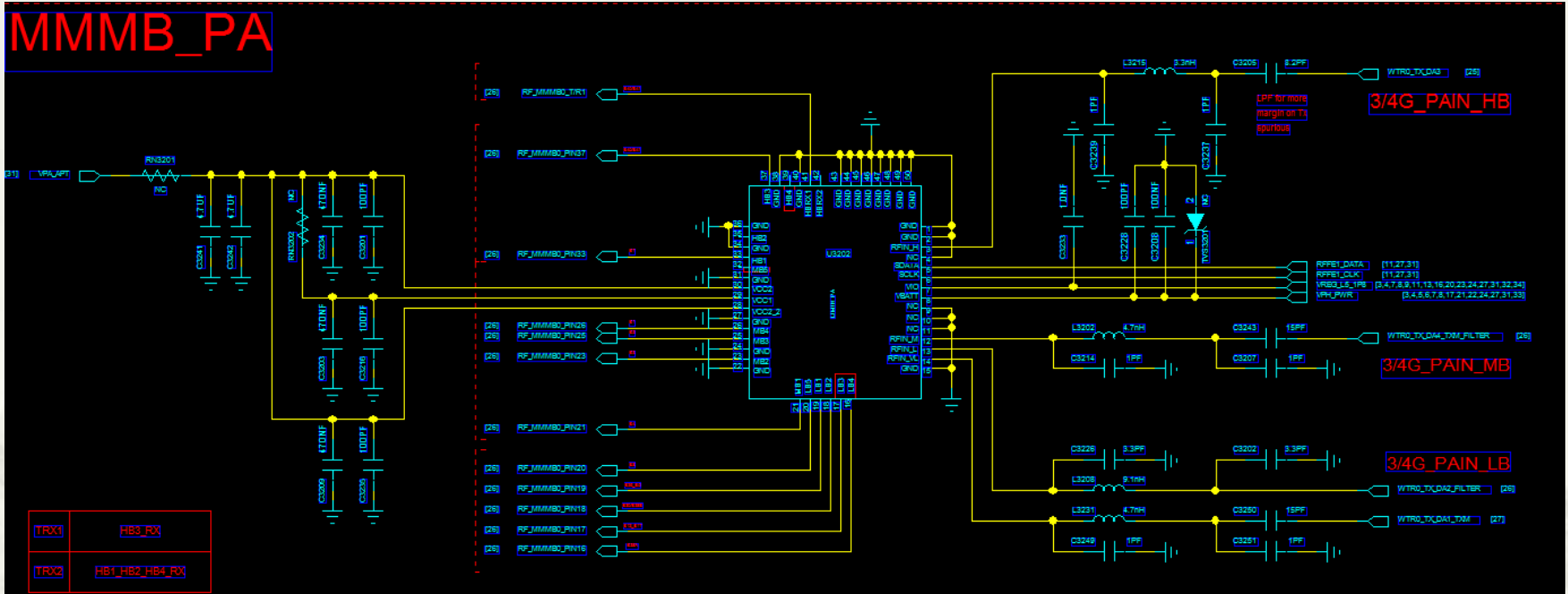
The output wave of Crystal looks like sine wave with amplitude 1V, frequency 19MHz



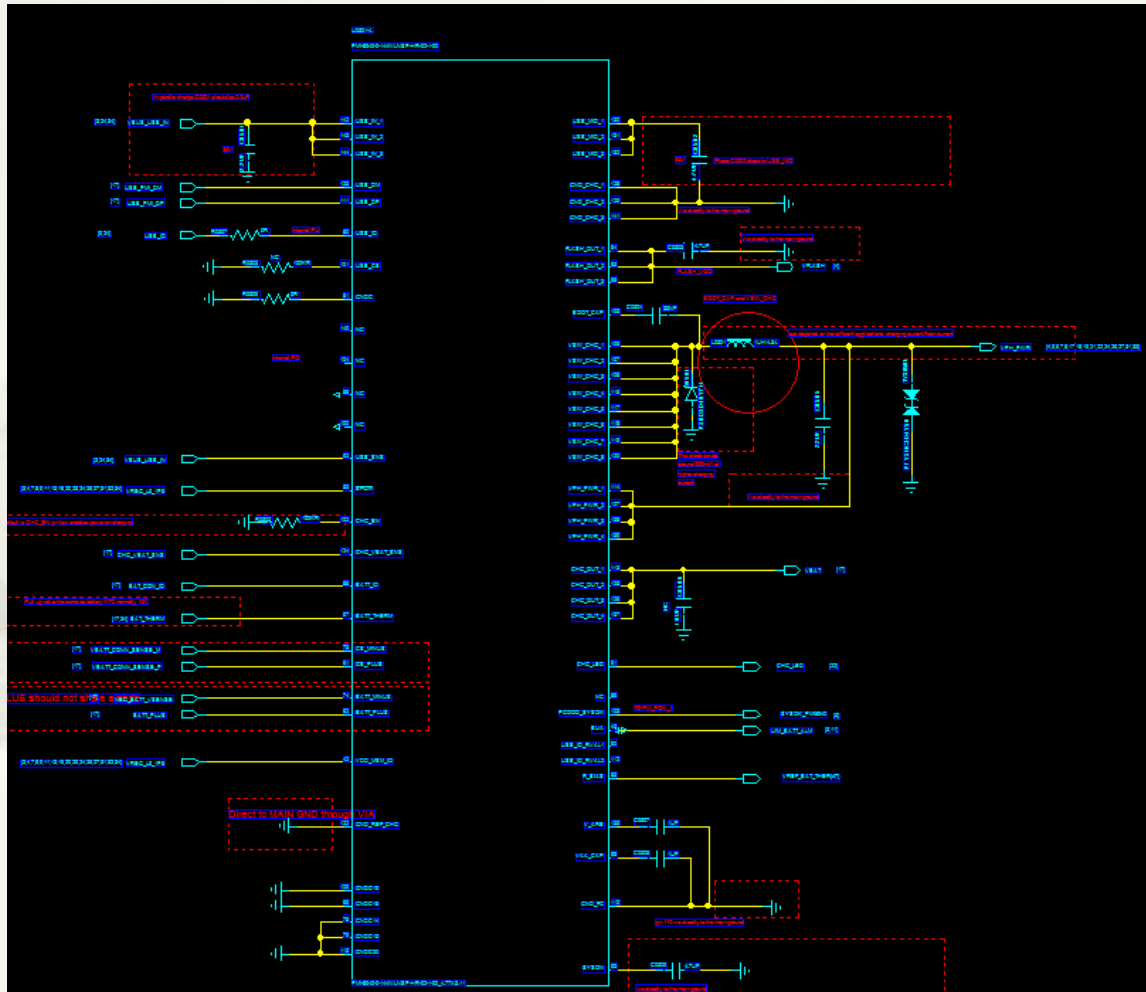
SP8T U3600



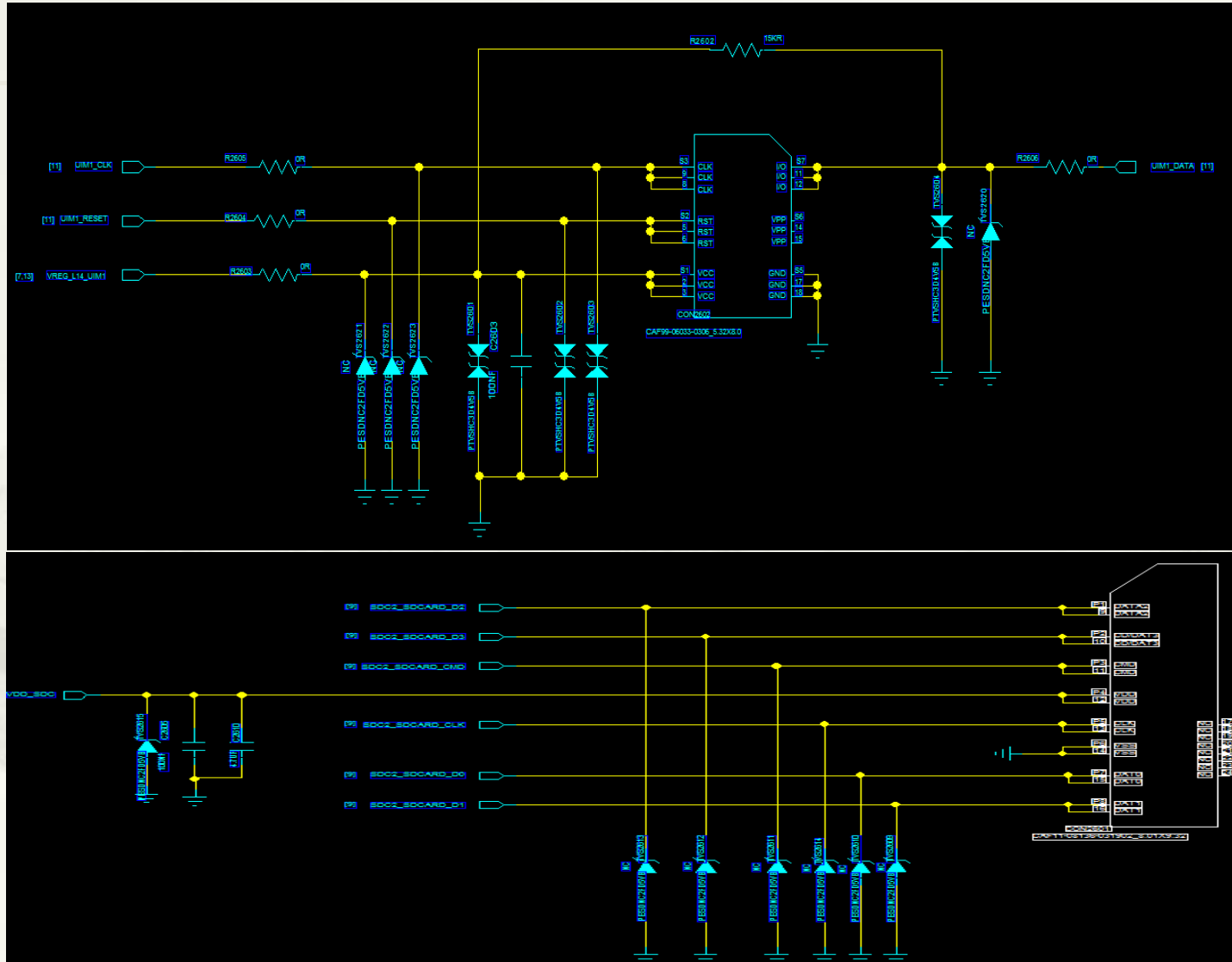
MMPA U3202



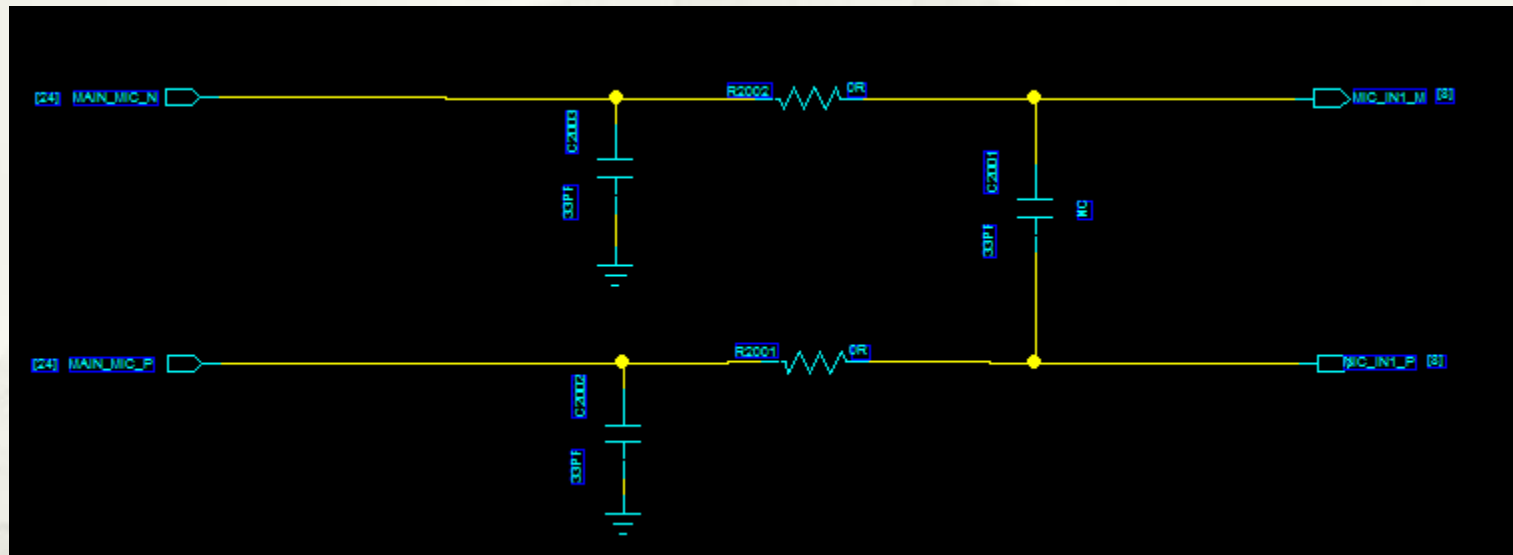
Charge management PMI8940 U0501



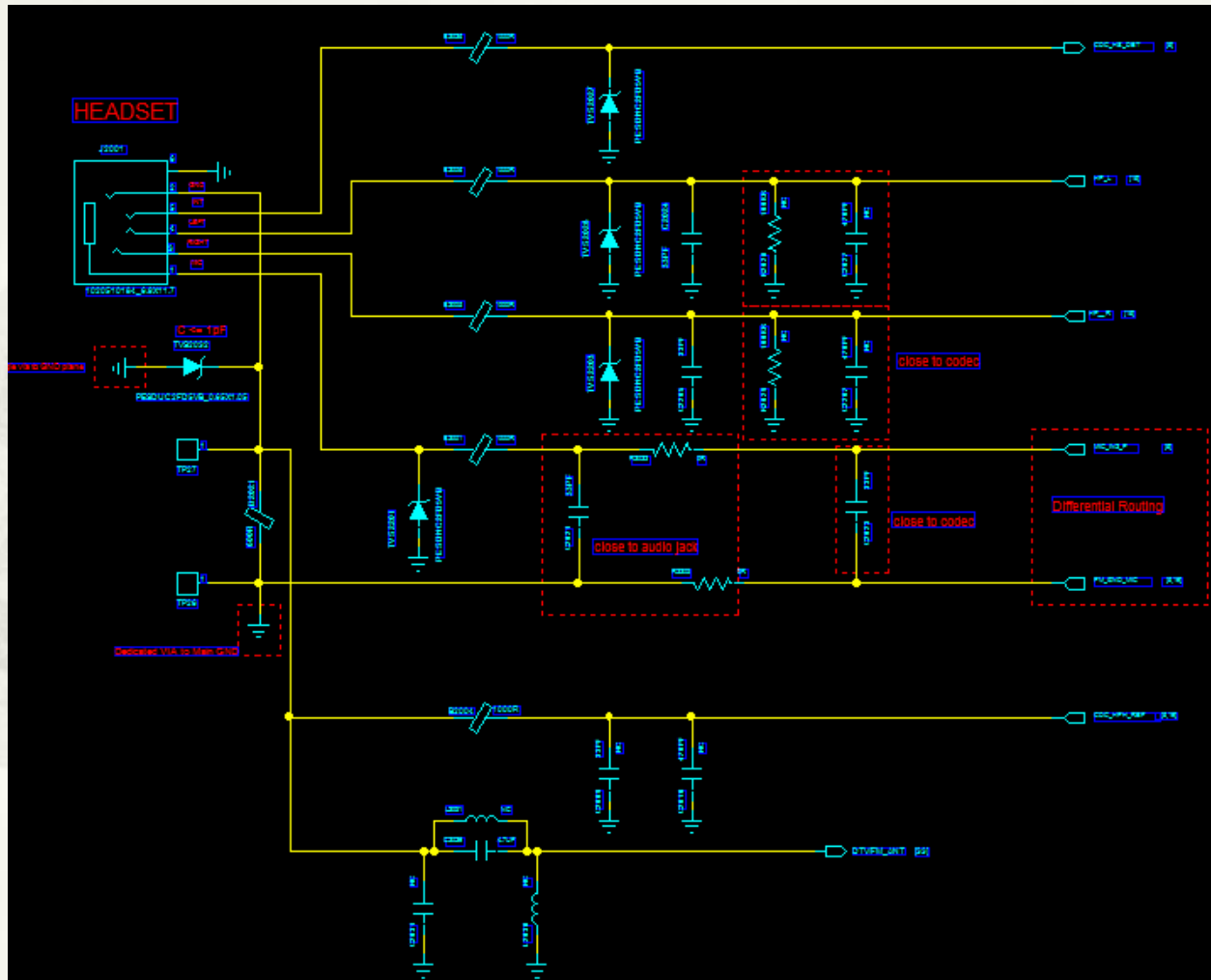
SIM & TF socket con2602 & con2603 con2601



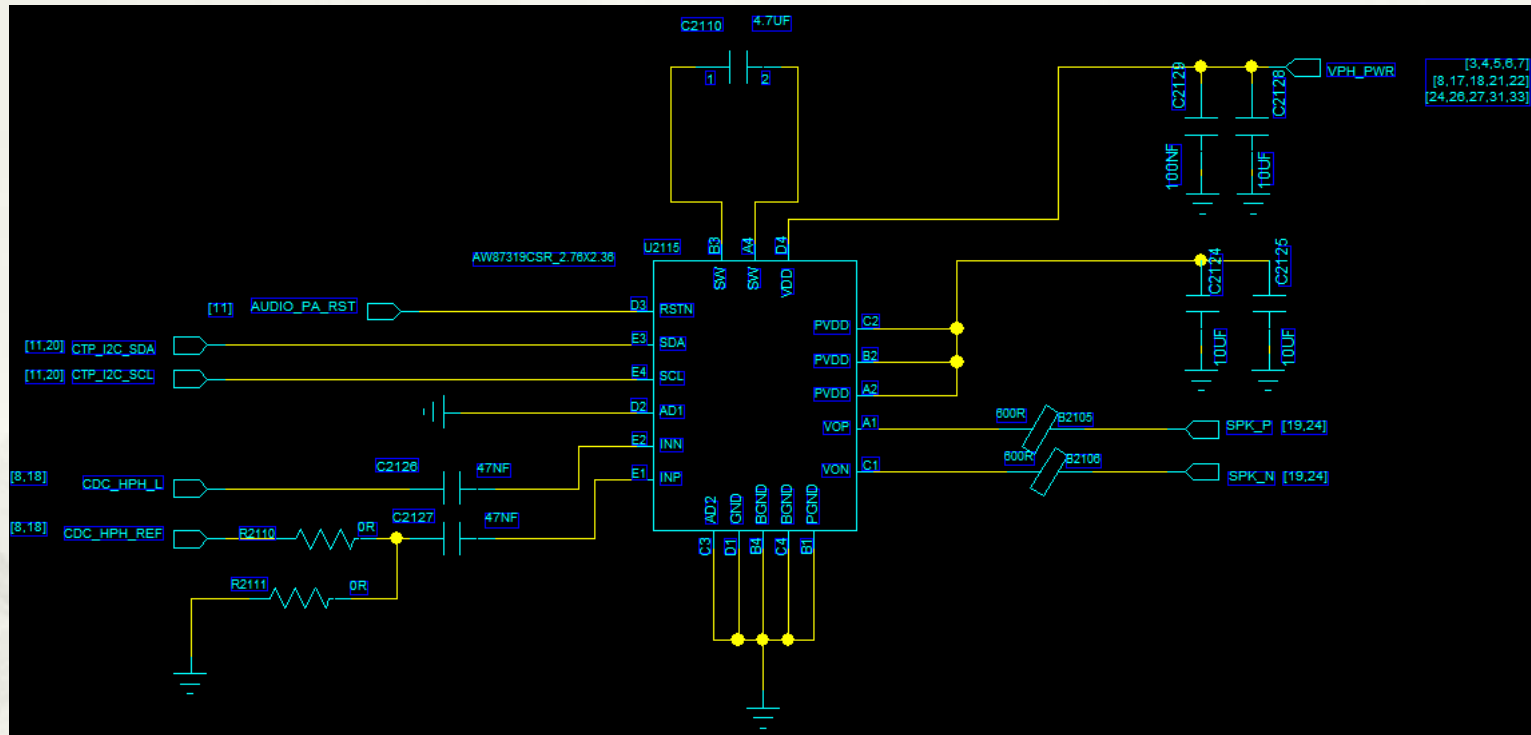
Main MIC



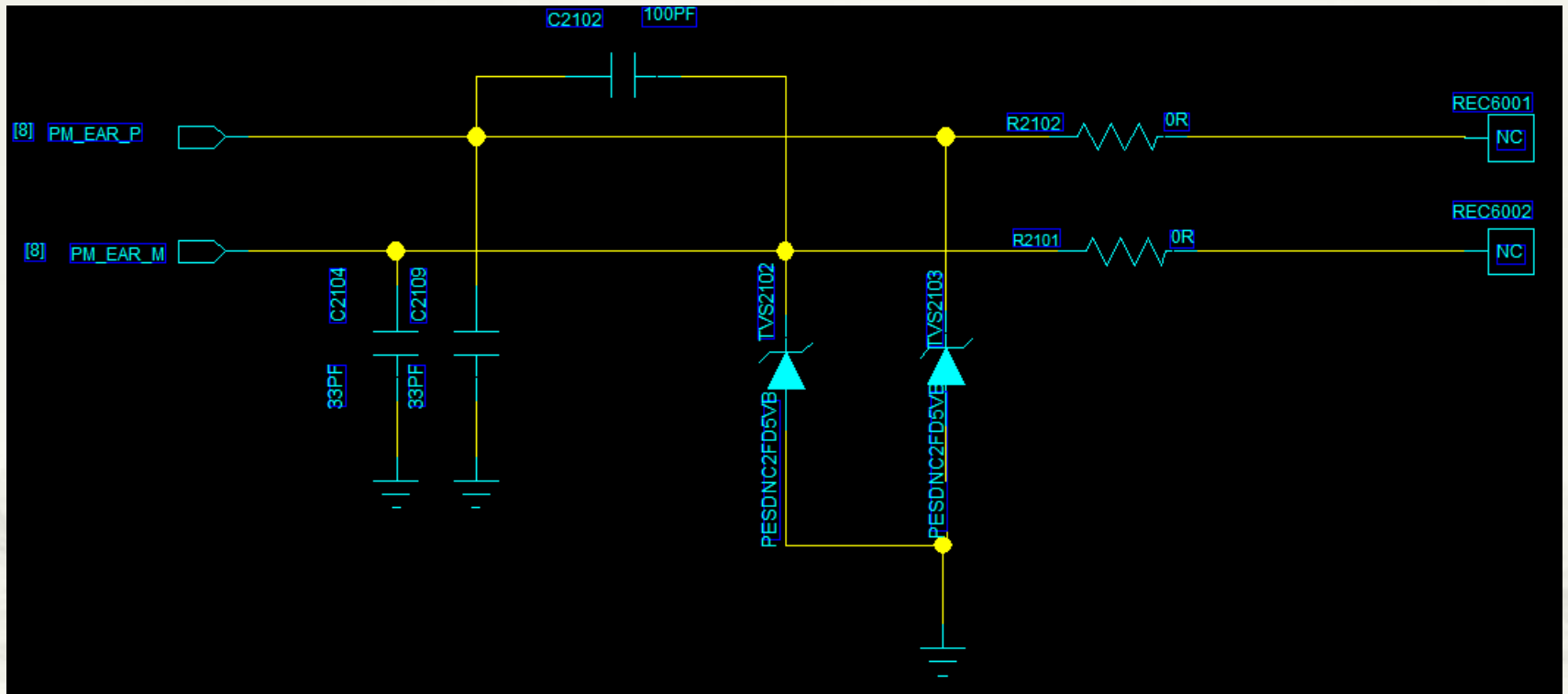
Headset Jack section J2001



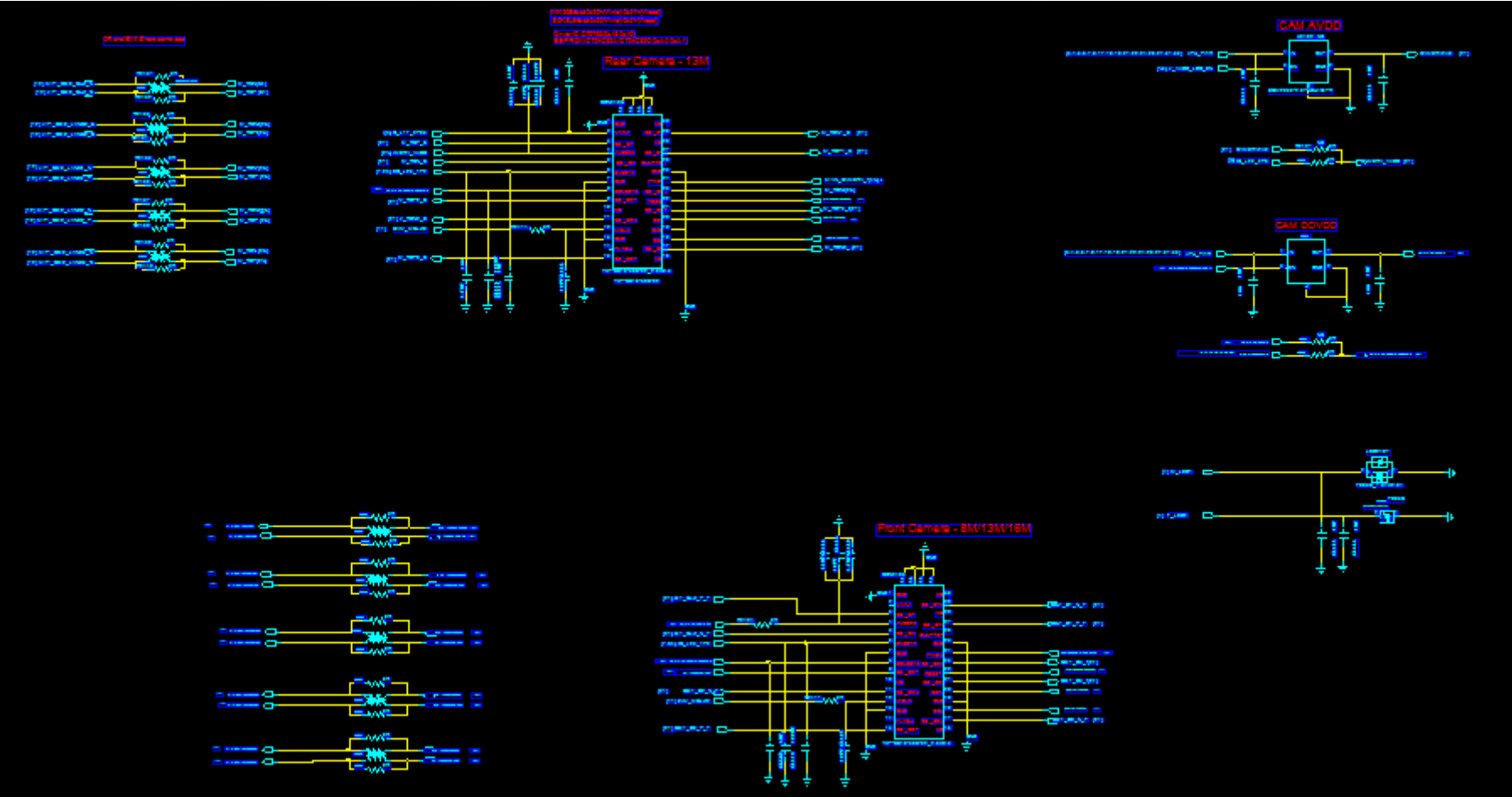
Speaker PA AW87329 U2115



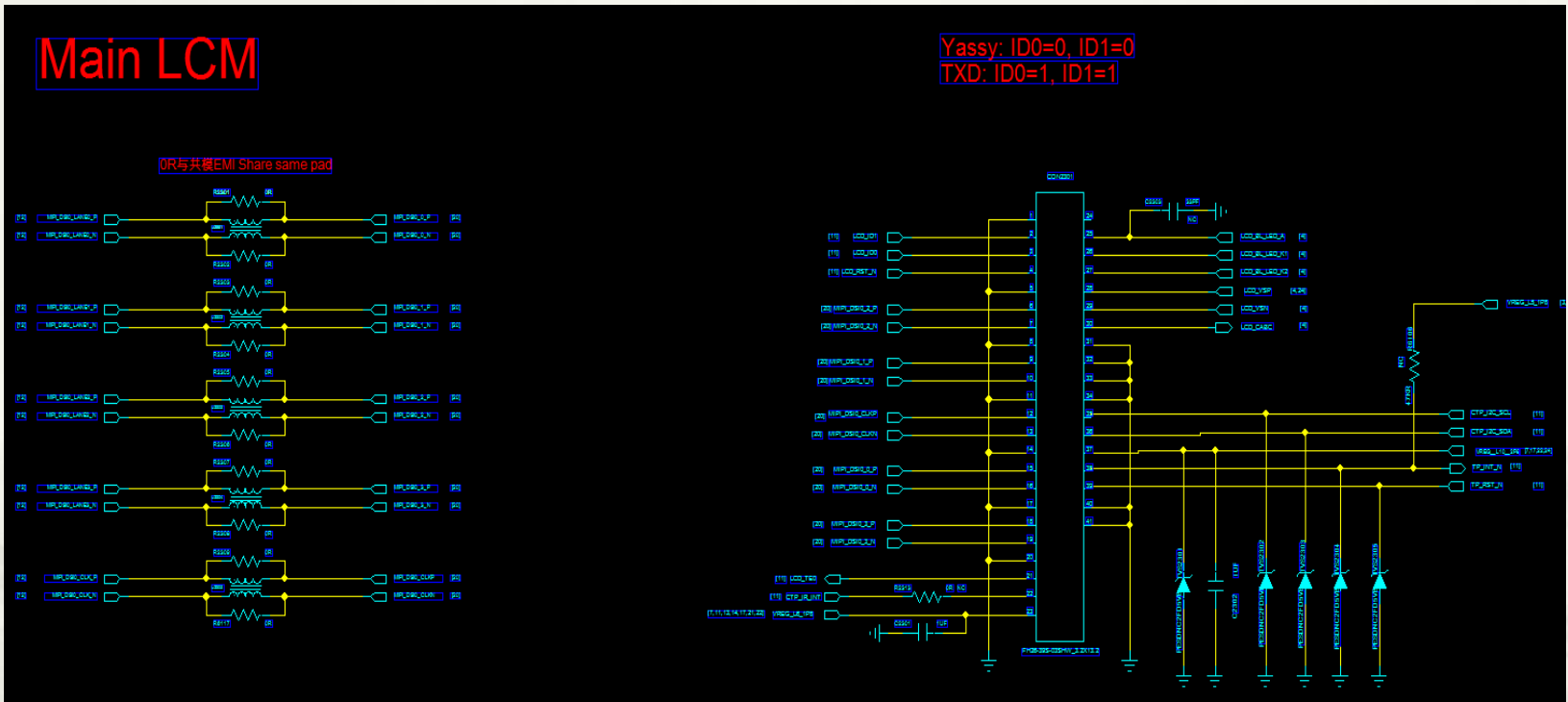
Receiver section



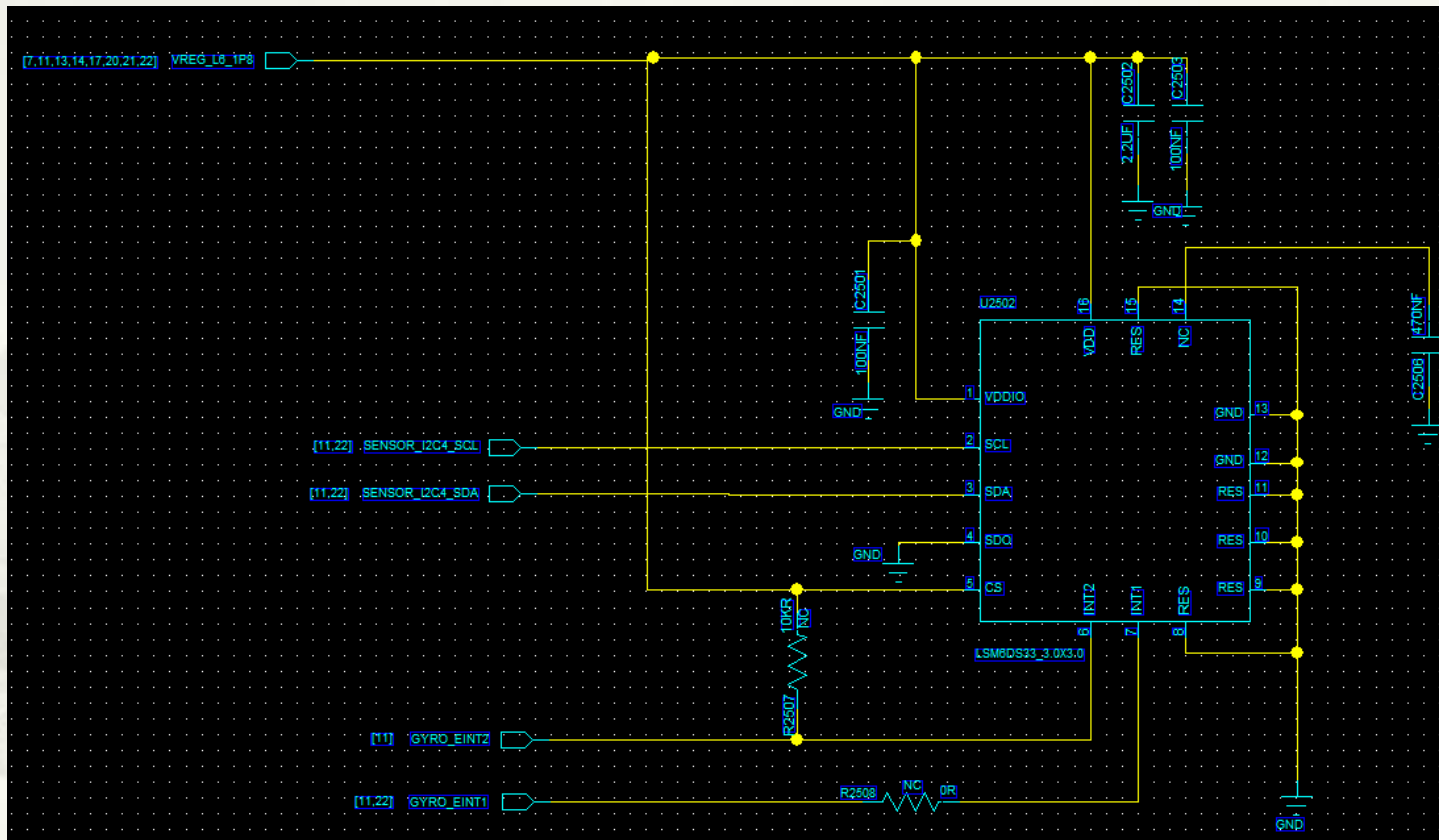
Camera section



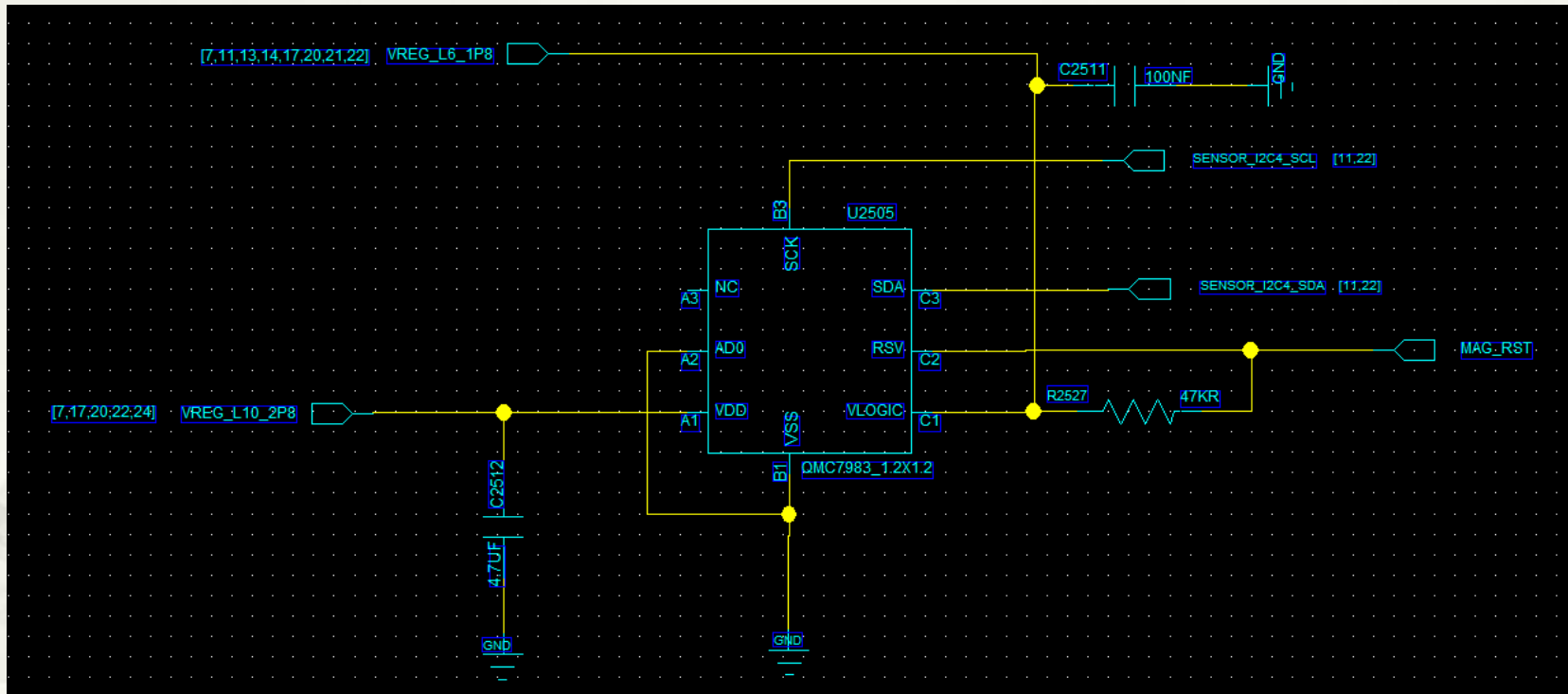
LCM section con2301



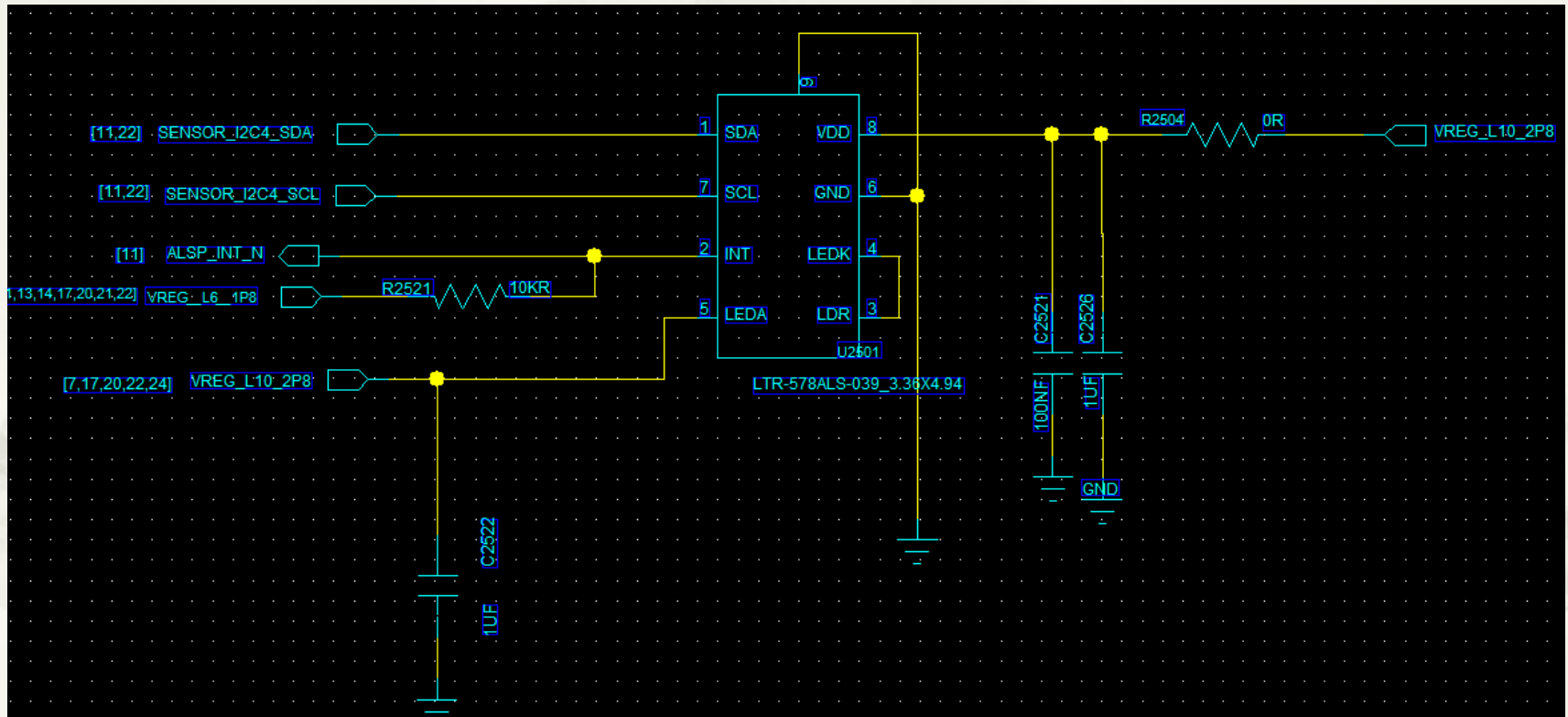
G-sensor section U2502



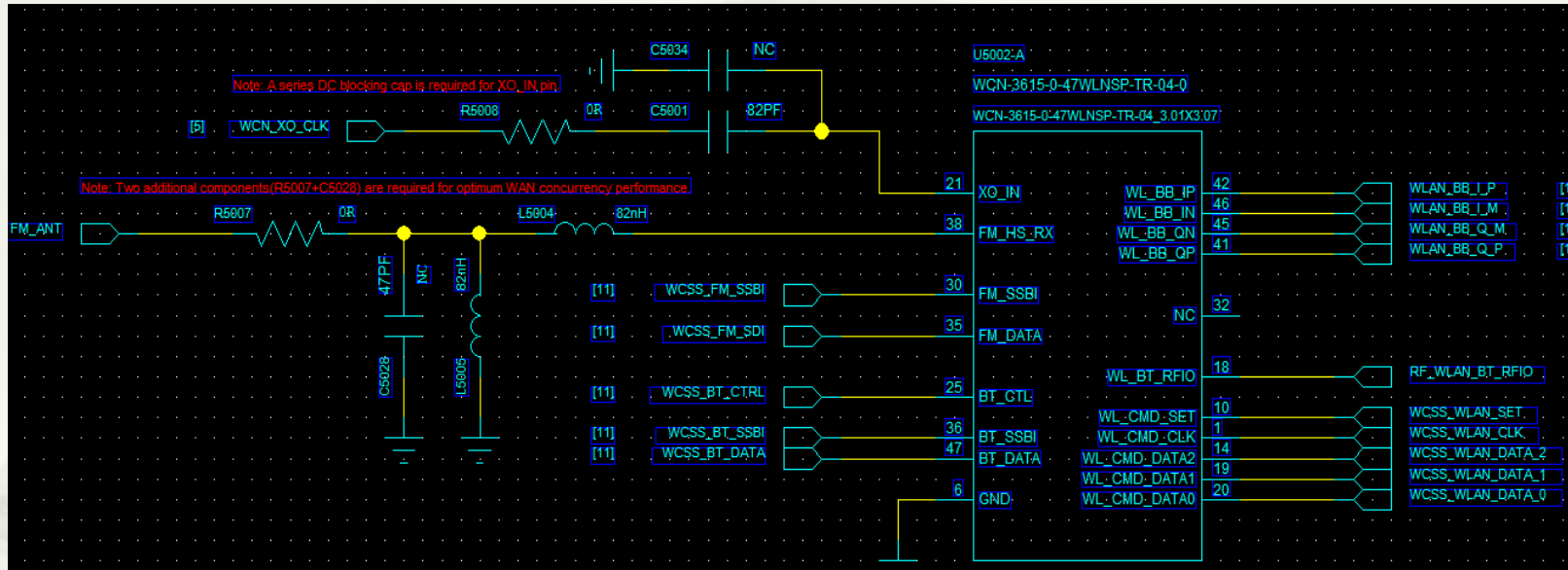
M-sensor section U2505



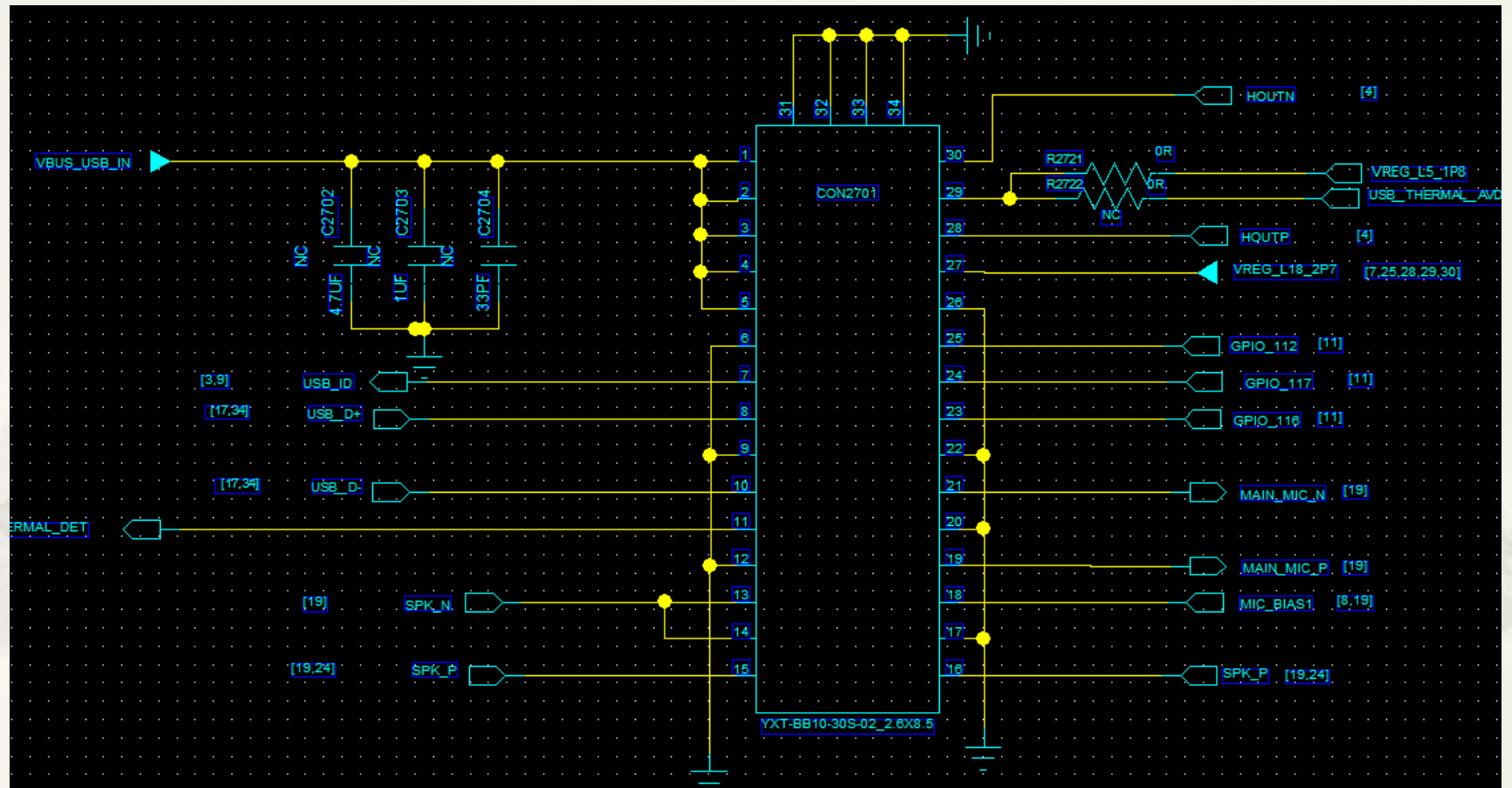
Ambient Light Sensor & Proximity Sensor section U2501



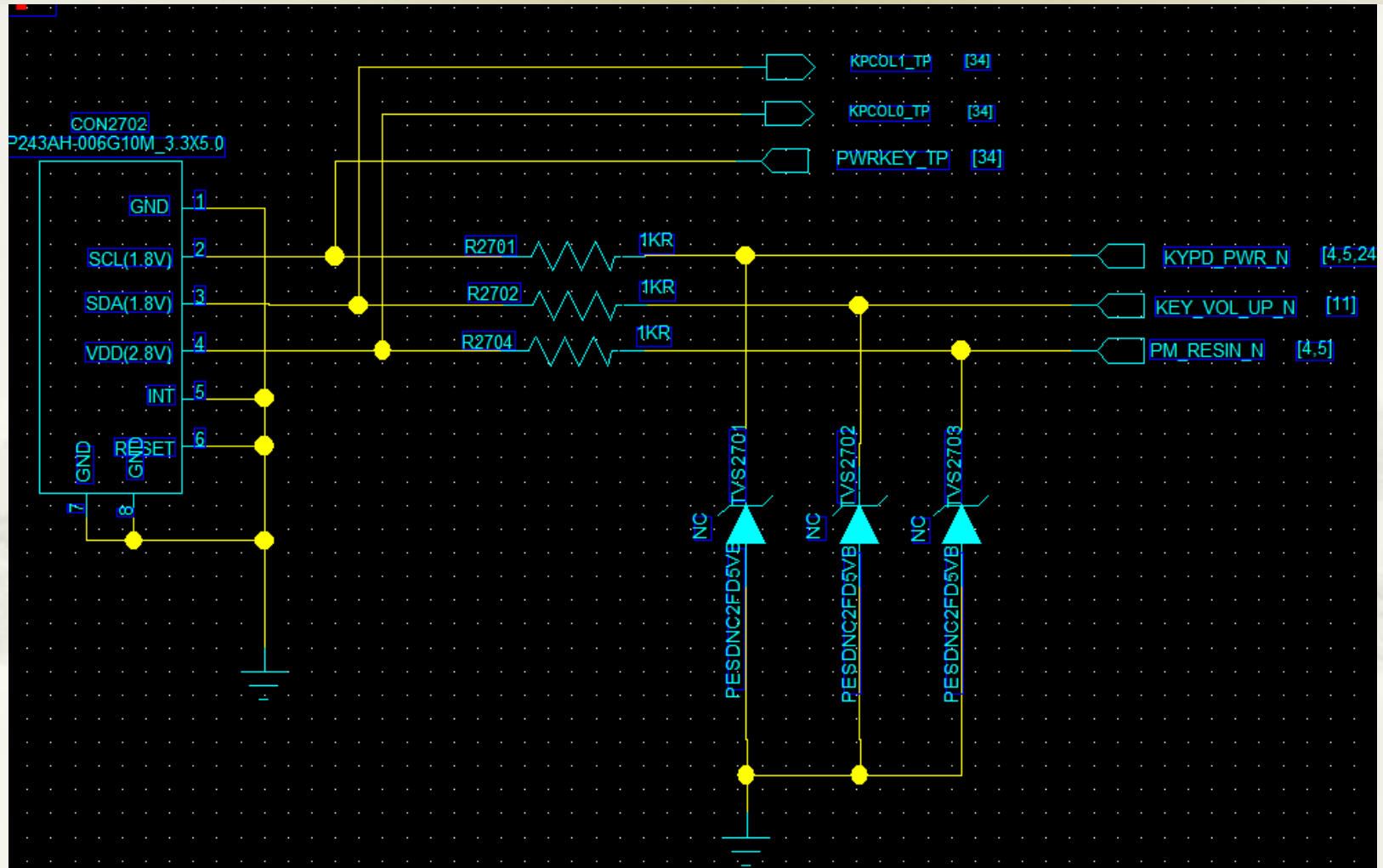
GPS/WIFI/FM/Bluetooth WCN3615



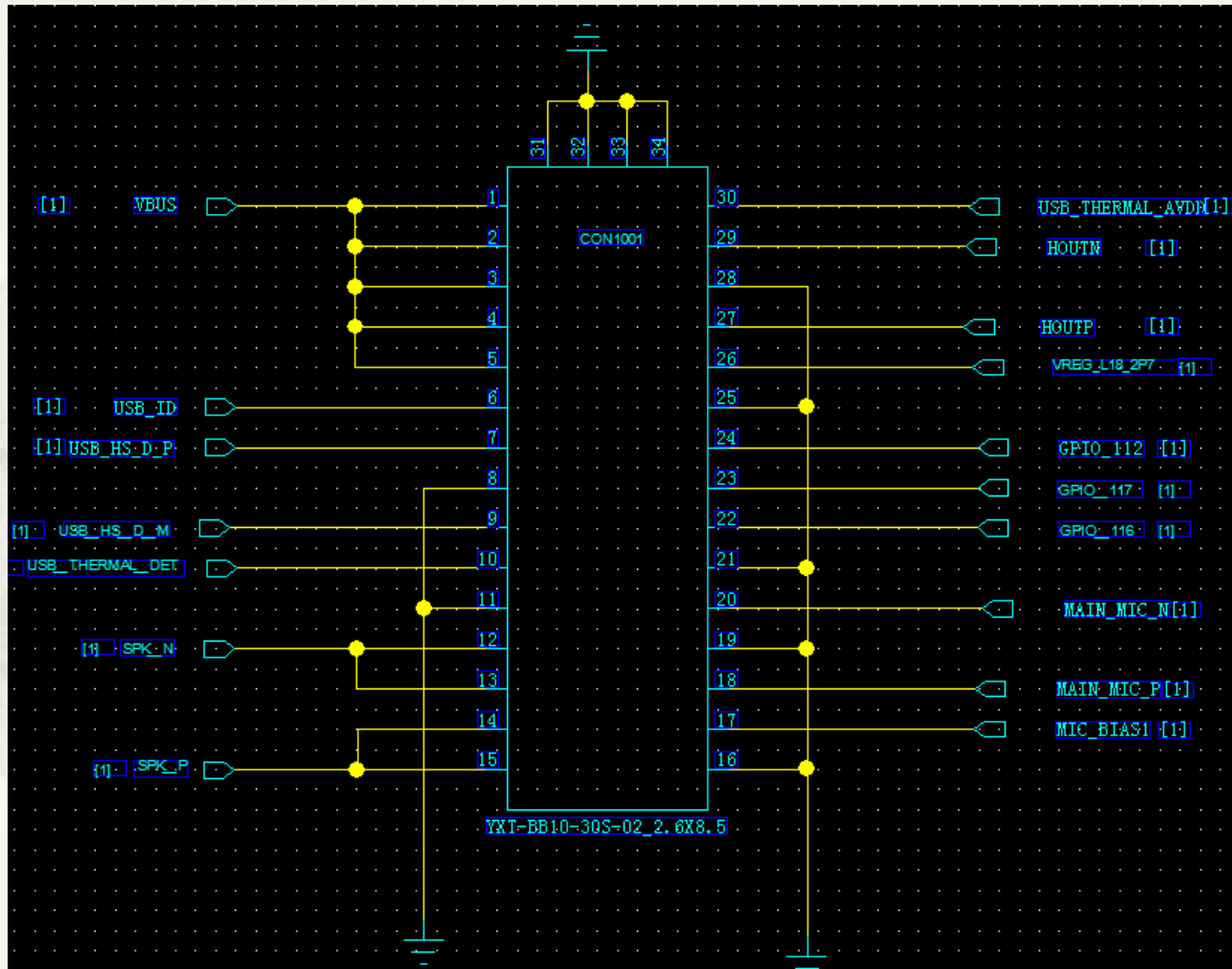
To sub board CON2701



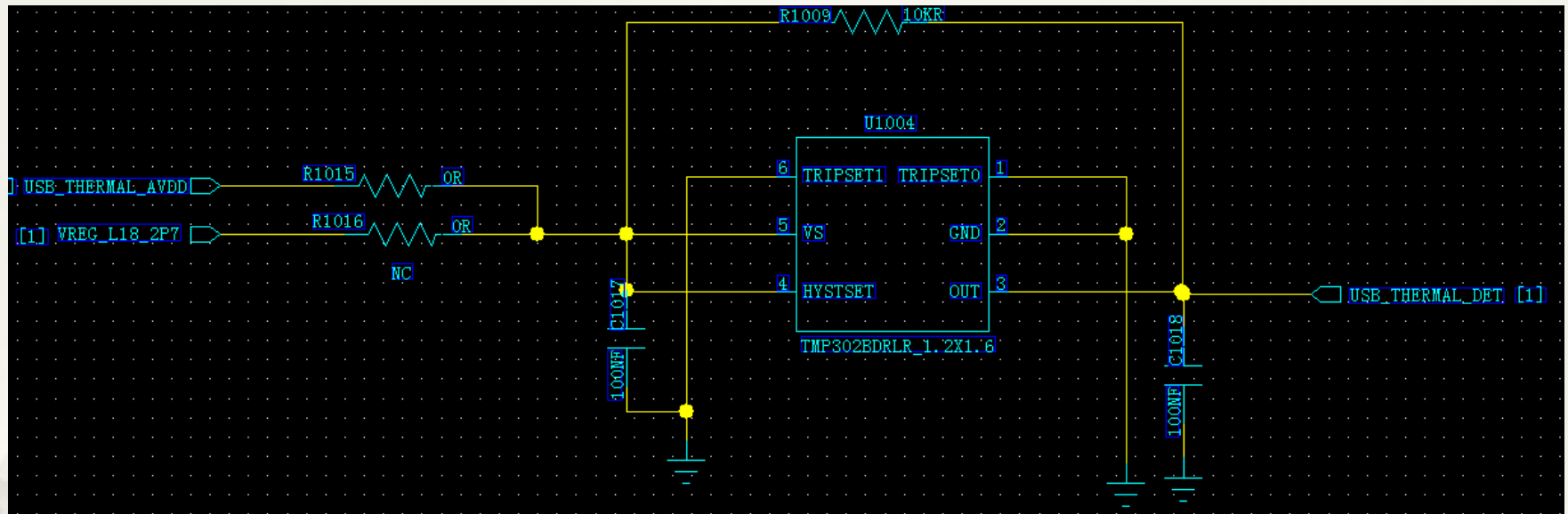
Side key section CON2702



Sub board



Thermal Alert



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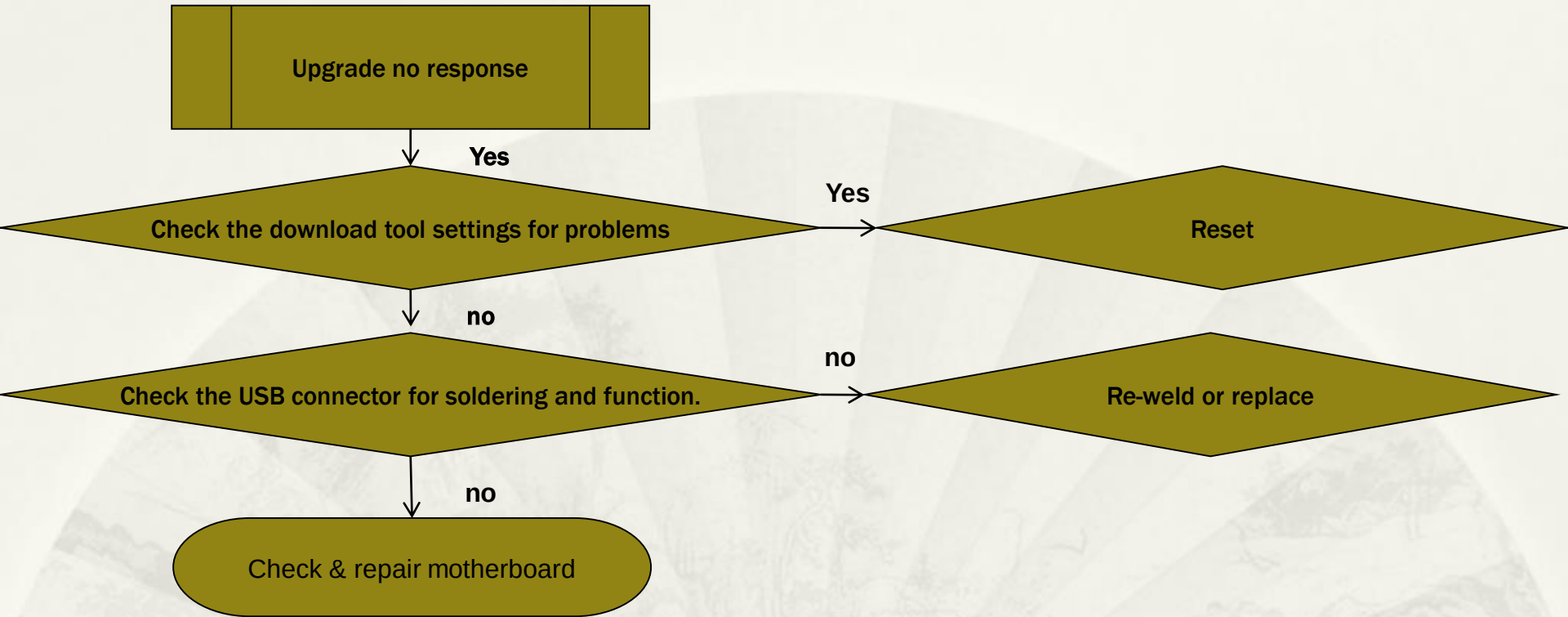
System Block Diagram

I2C address and power management

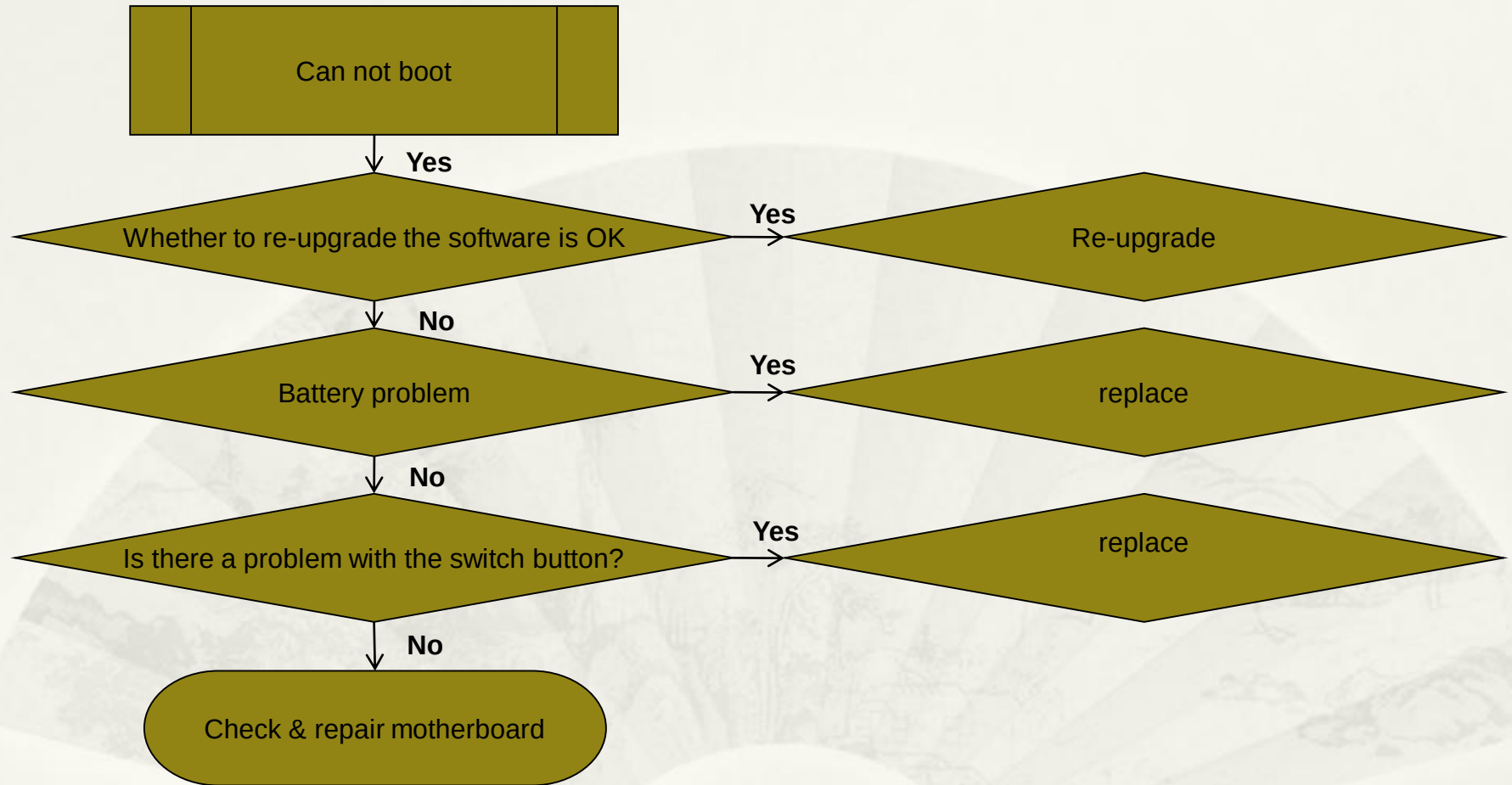
Important devices and key test points

Maintenance method

1. Upgrade failure analysis



2. Can not boot fanalysis



2.1 Can not boot fanalysis _ power-on voltage and current abnormalities

- * Use a multimeter or oscilloscope to measure the battery cell P+ voltage is low, and the power is too low, the voltage is lower than 3.4V:
 - Charging, after the voltage is greater than 3.6V, re-execute the boot action.
- * The battery cell P+ voltage is greater than 3.6V, but there is leakage current at power-on:
 - Check if TVS1901 is abnormal;
 - Check if the charging management chip U0501 is abnormal;
 - Confirm whether the RF PA U3301, U3202 is abnormal;
 - Check if U2501, U2502, U2505 are abnormal;
 - Check if the power management chip U0701 is abnormal;
 - Check if there is a short circuit in CON2101 pin1;
 - In addition to the PMIC U0701, the above devices do not affect the booting. The analysis can be removed one by one and then restarted.

2.2 Can not boot analysis - boot process current abnormal

- * The VBAT voltage is high with a multimeter or an oscilloscope, but the current is abnormal during boot.
- Check if there is a chip fever
- Check if the voltage output of each PMIC is normal
- Check if the CPU is abnormal

2.3 Can not boot analysis - clock

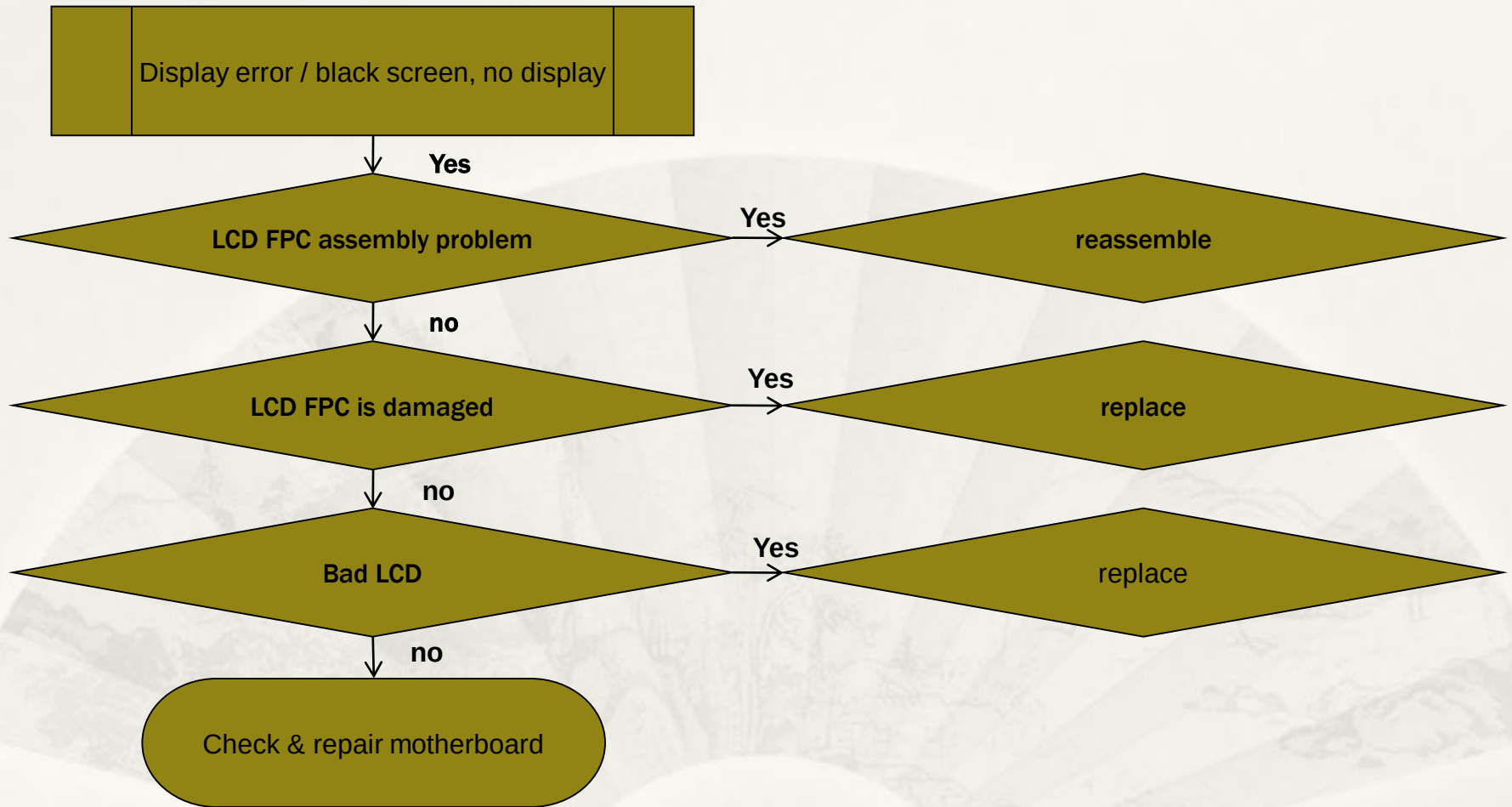
- * VBAT voltage is high with a multimeter or oscilloscope
- Check if there is any abnormality in the output voltage of each PMIC U0701. If there is any abnormality, add the welding or replace the PMIC for further confirmation. If there is no abnormality, proceed to the next step.
- Check if the clock X0701 output clock is normal
- Check if the 19.2Mhz clock U3102 output 26M clock is normal

2.4 Can not boot analysis eMMC

--Confirm eMMC monomer U1801



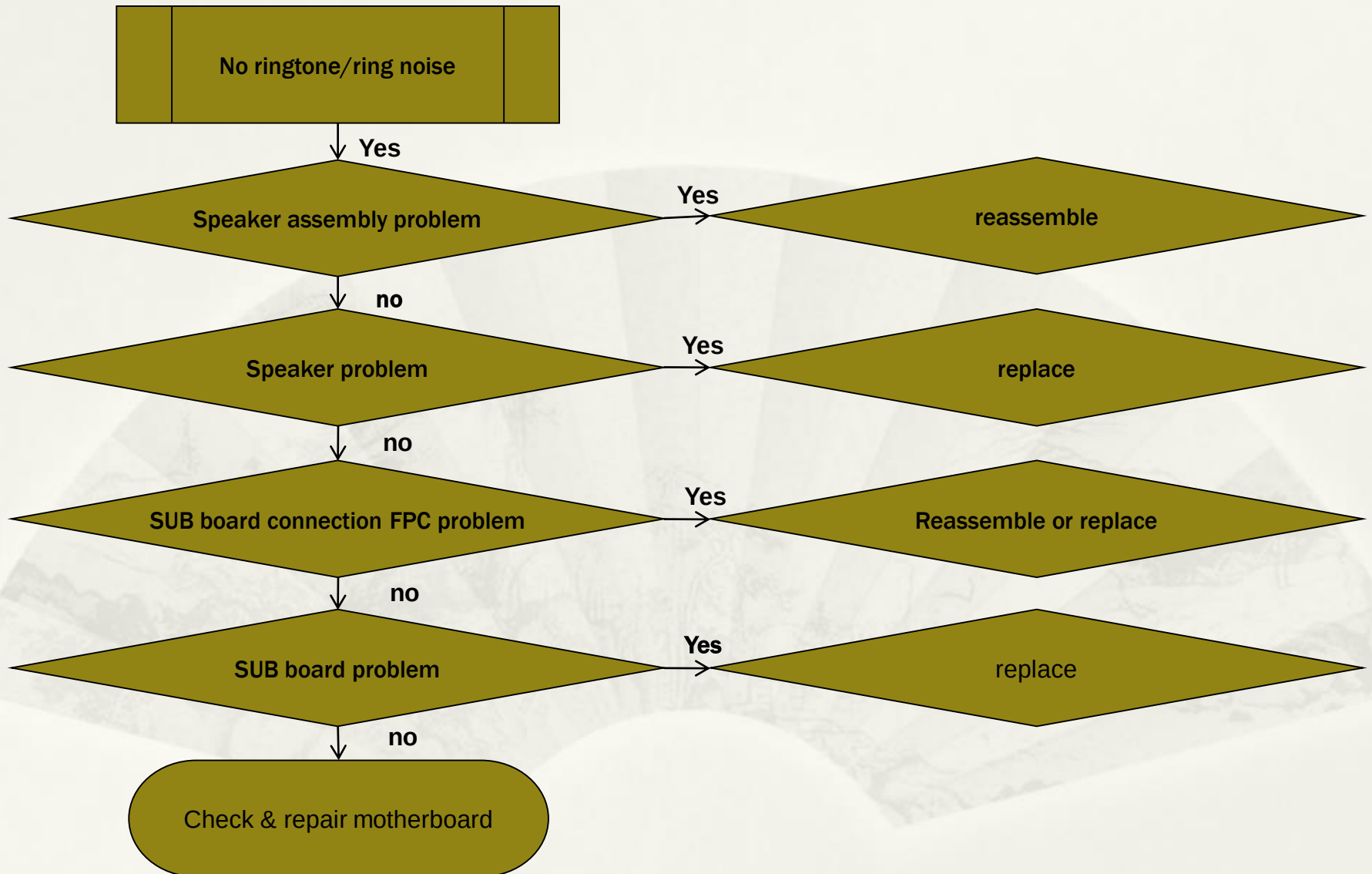
3. Display abnormal fault analysis



3.1 Display abnormal fault analysis LCM

- Confirm that CON2301 has open pad or damage.
- It has been confirmed that the power supply VIO28_PMU and VIO18_PMU are normal after the boot is completed.
- Check if the backlight is normally lit. If the backlight is not bright, check if the backlight chip U0501 and its peripheral devices are normal. When the backlight is bright, proceed to the next step.
- Measure whether all signals are on the R6108-R6115. If not all signals, confirm the appearance of the CPU U1101 and X-ray. If there is any abnormality, it can be soldered or replaced.

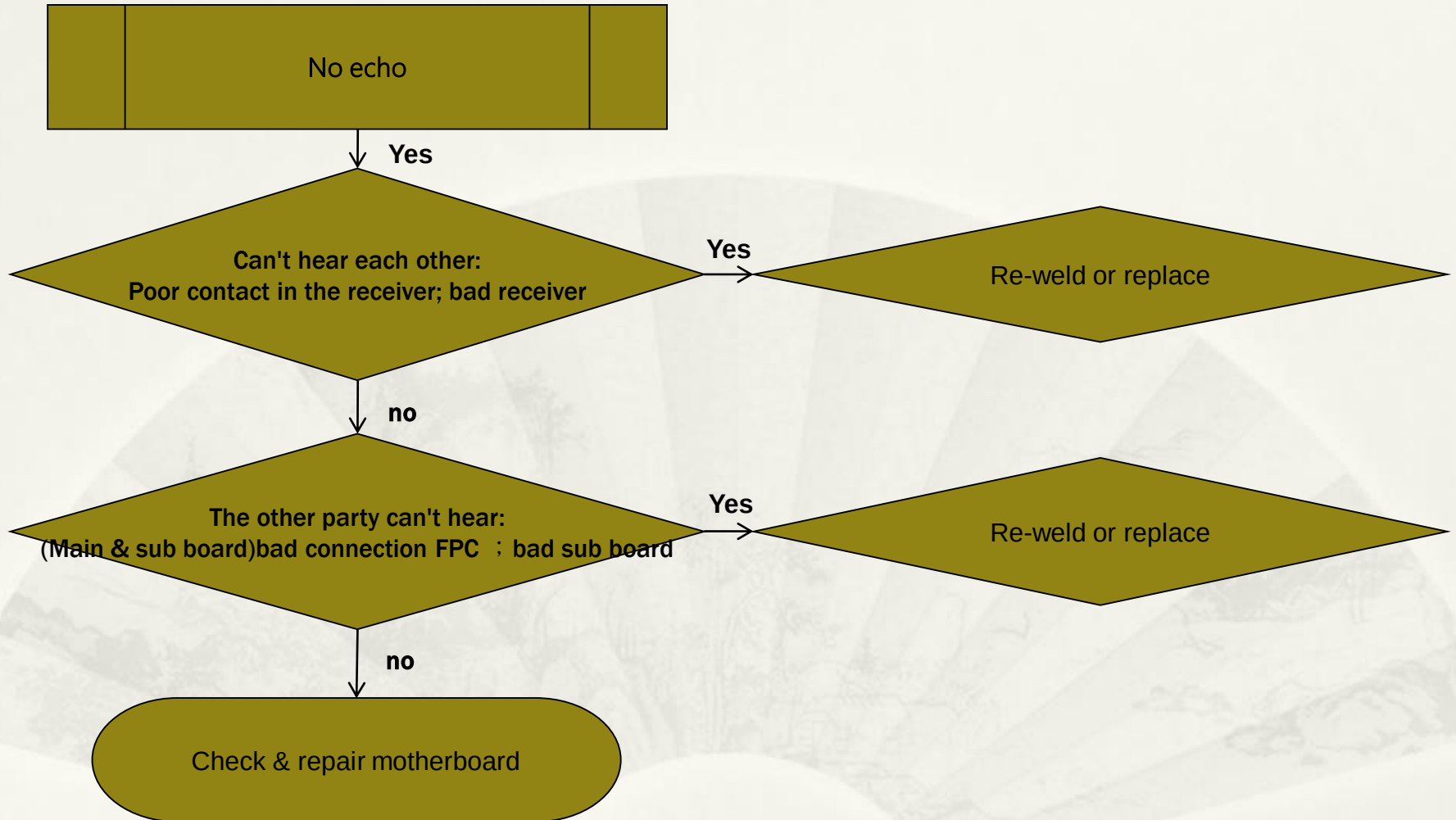
4. Ringtone failure analysis



4.1 Ringtone failure analysis

- Check if B2109, B2110, C2111, C2112, C1001, C1003 are normal, and re-weld or replace if abnormal.
- Check if the main and sub board connectors CON2701 have open pad or damage parts
- Confirm the appearance of the PMIC U0701 and X-ray, no abnormalities can be added or further replaced

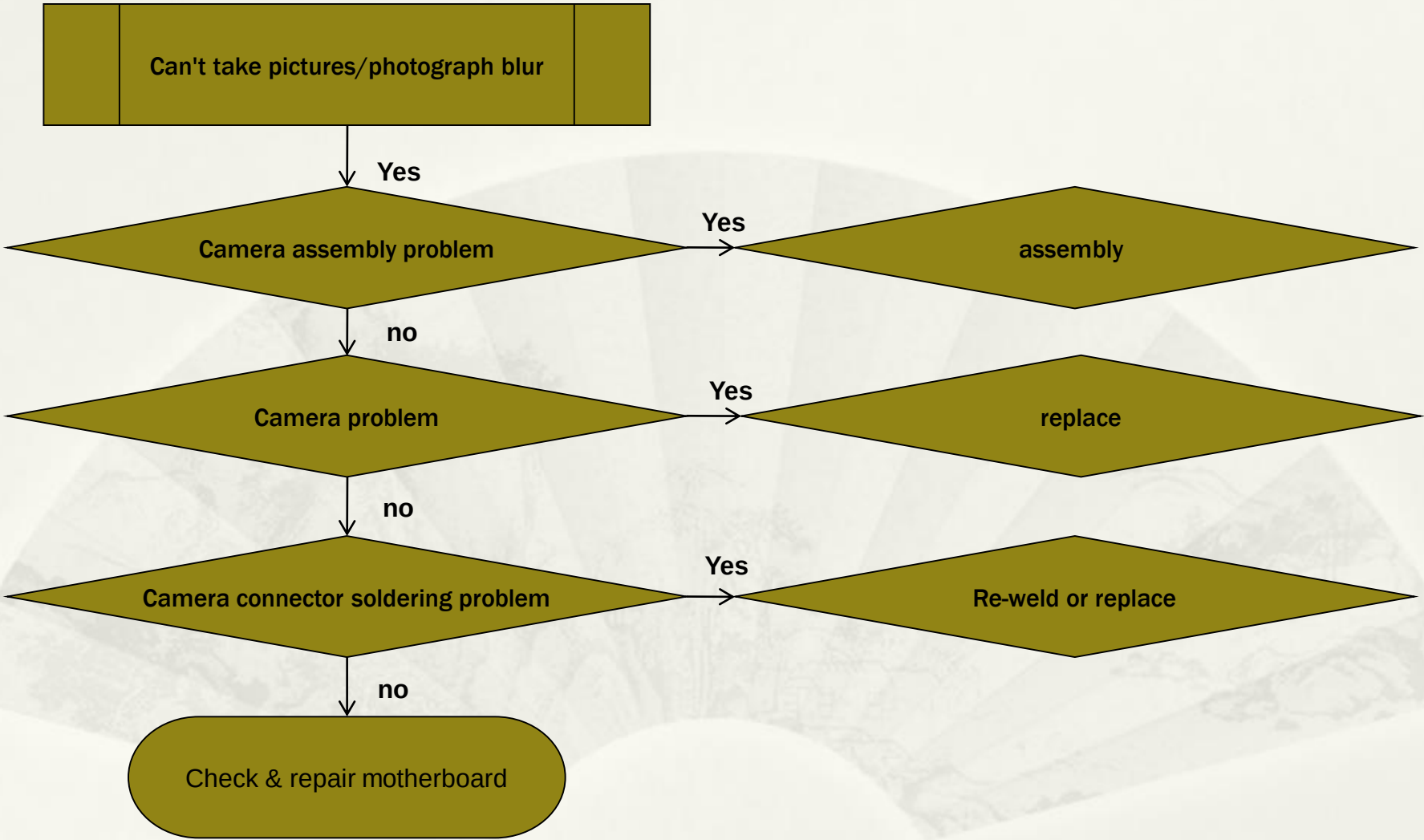
5. No echo failure analysis



5.1 No echo failure analysis

- Confirm whether the materials around the earpiece C2104, C2109, C2102, R2102, R2101, TVS2102, TVS2103 have air welding or damage.
- Confirm whether the sub board main MIC materials C101, R1010, R1011, TVS102, C1007, C1013, C1014 have empty welding or damage.
- Confirm whether the main and sub board connectors CON1001 pin17, 18, 20 CON2701 pin18, 19, 21 are soldered and the connector is damaged.
- Confirm the appearance of the PMIC U0701 and X-ray, no abnormalities can be added or further replaced

6. Bad photo Failure analysis



6.1 Bad photo failure analysis

-- Confirm that the main camera connector CON2402, the rear sub-camera CON2401, and the front camera connector CON2403 are open pad or damaged according to the main and sub-photo defects.

-- Confirm that the power supply is normal.

-- Confirm that the main-short circuit module R2431-R2440 is normal.

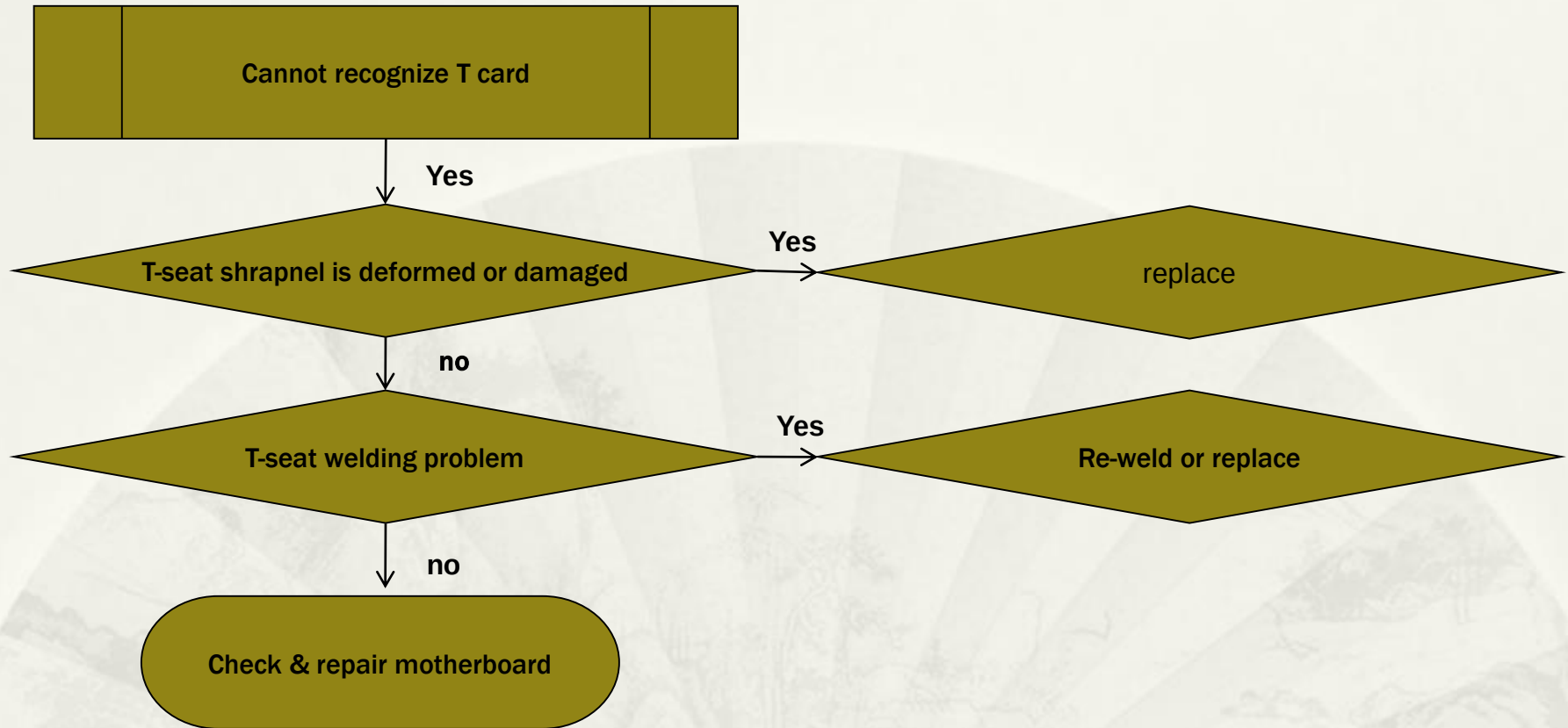
After confirming, is the sub-short circuit module R2415-R2442 normal?

Confirm that the front-short circuit module R2443-R2452 is normal.

--Check if the camera switch U2401 is abnormal

-- Confirm U0701 appearance and X-ray, no abnormalities can be welded or further replaced

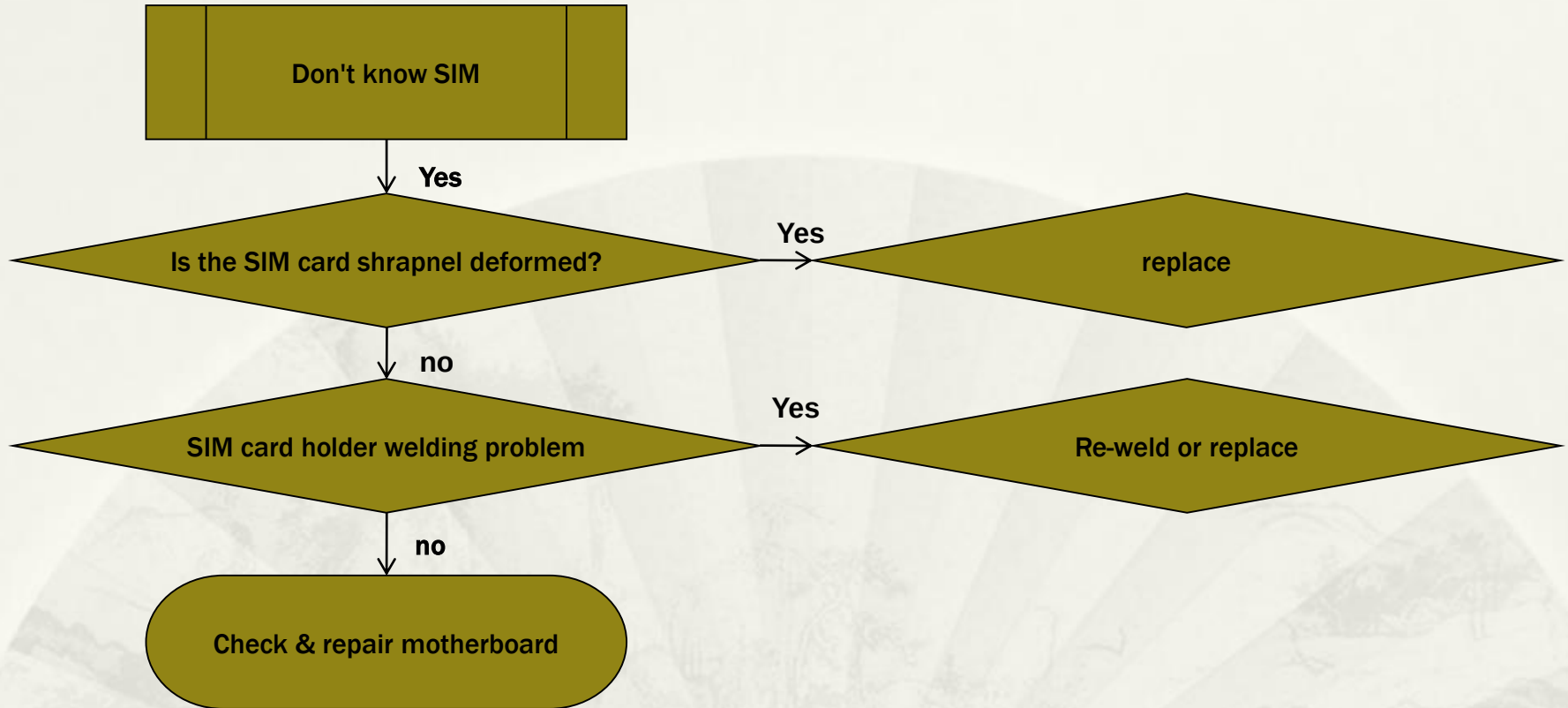
7.T Card failure analysis



7.1 T card failure analysis

- Confirm whether the relevant TVS tube has empty welding or damage.-- Check if the power supply is normal, whether the VDD_SDC output is 3.0V, confirm C2605, C2610 ◦
- Confirm that the interrupt EINT_CARD is normal, CON2601 and peripheral devices.
- Confirm the appearance of the CPU U1101 and X-ray, no abnormalities can be added or further replaced

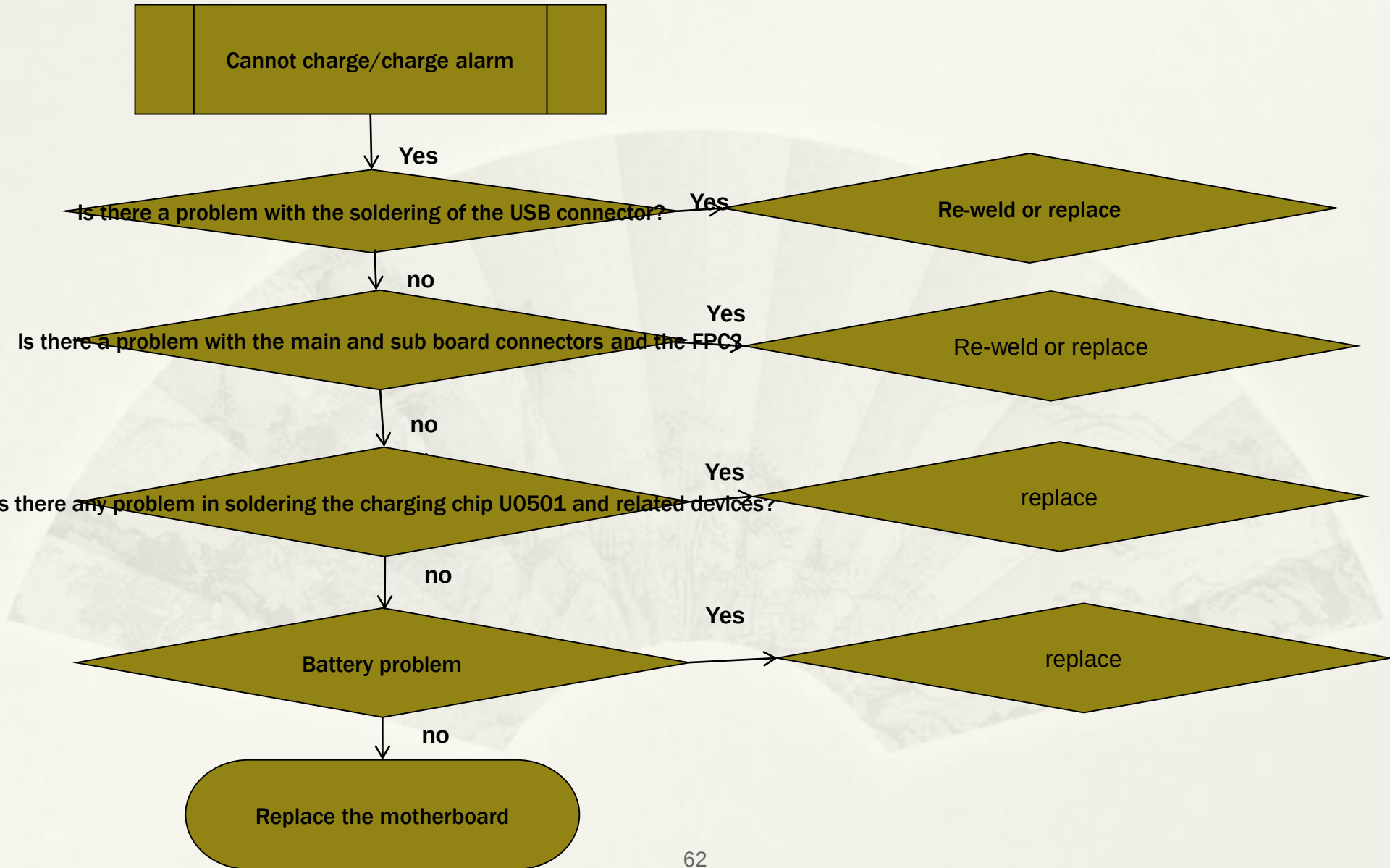
8. Do not know SIM failure analysis



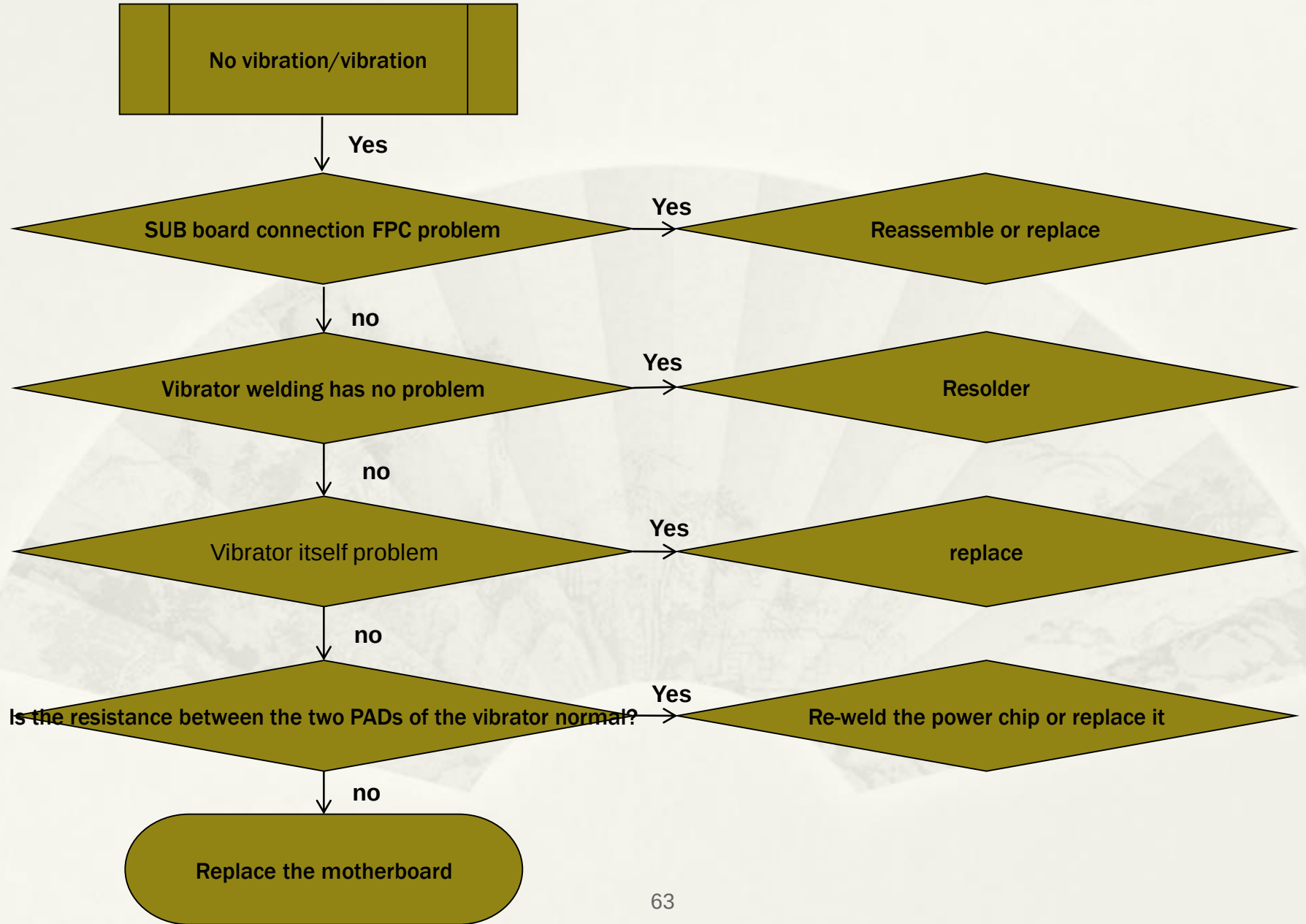
8.1 Do not know SIM failure analysis

- Check if the SIM card connector CON2602, CON2603 is deformed, welded, damaged
- Check if the power supply is normal, and whether the output of VREG_L14_UIM1 and VREG_L15_UIM2 is normal.
- Confirm interrupt UIM1_PRESENT, UIM2_PRESENT is normal
- Confirm the appearance of the CPU U1001 and X-ray, no abnormalities can be added or further replaced

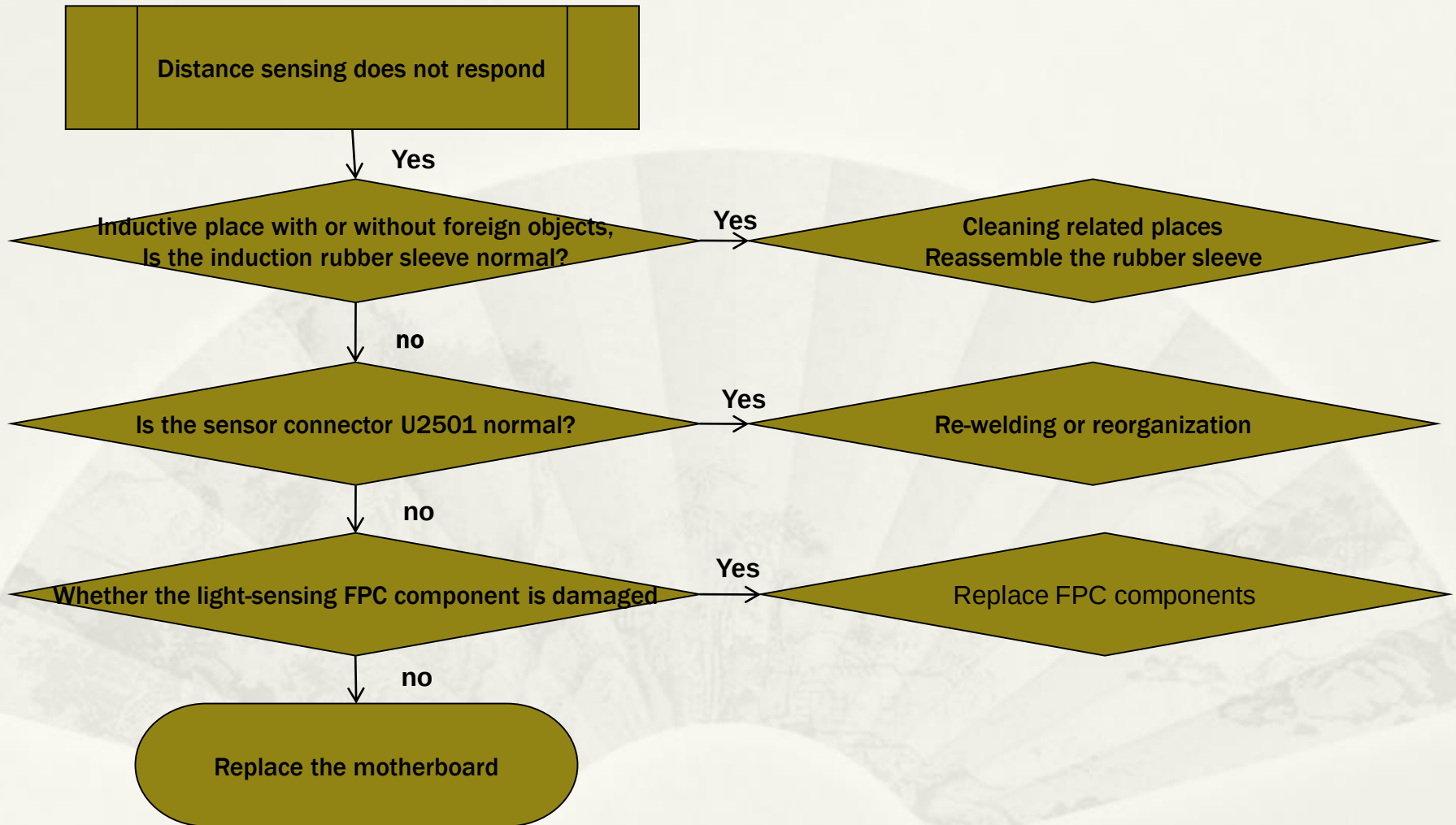
9. Charging failure analysis



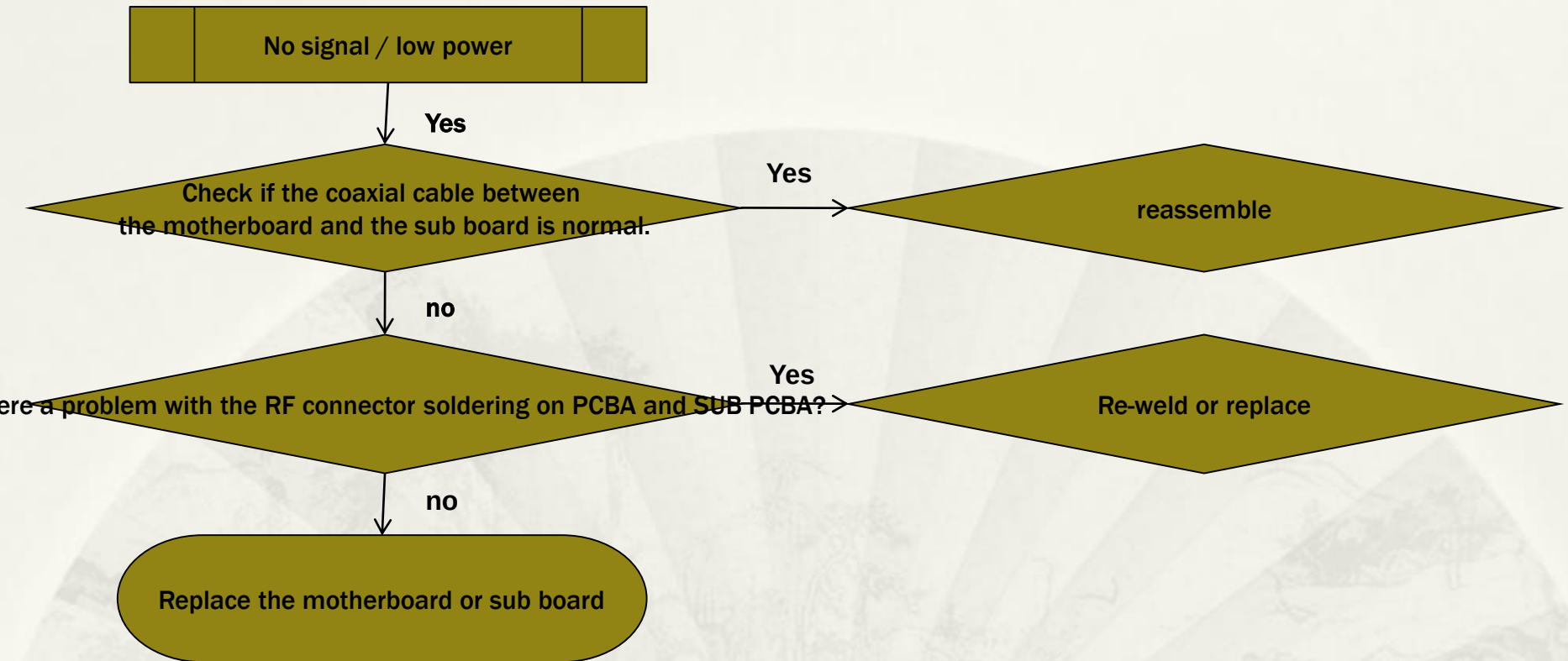
10. Vibration failure analysis



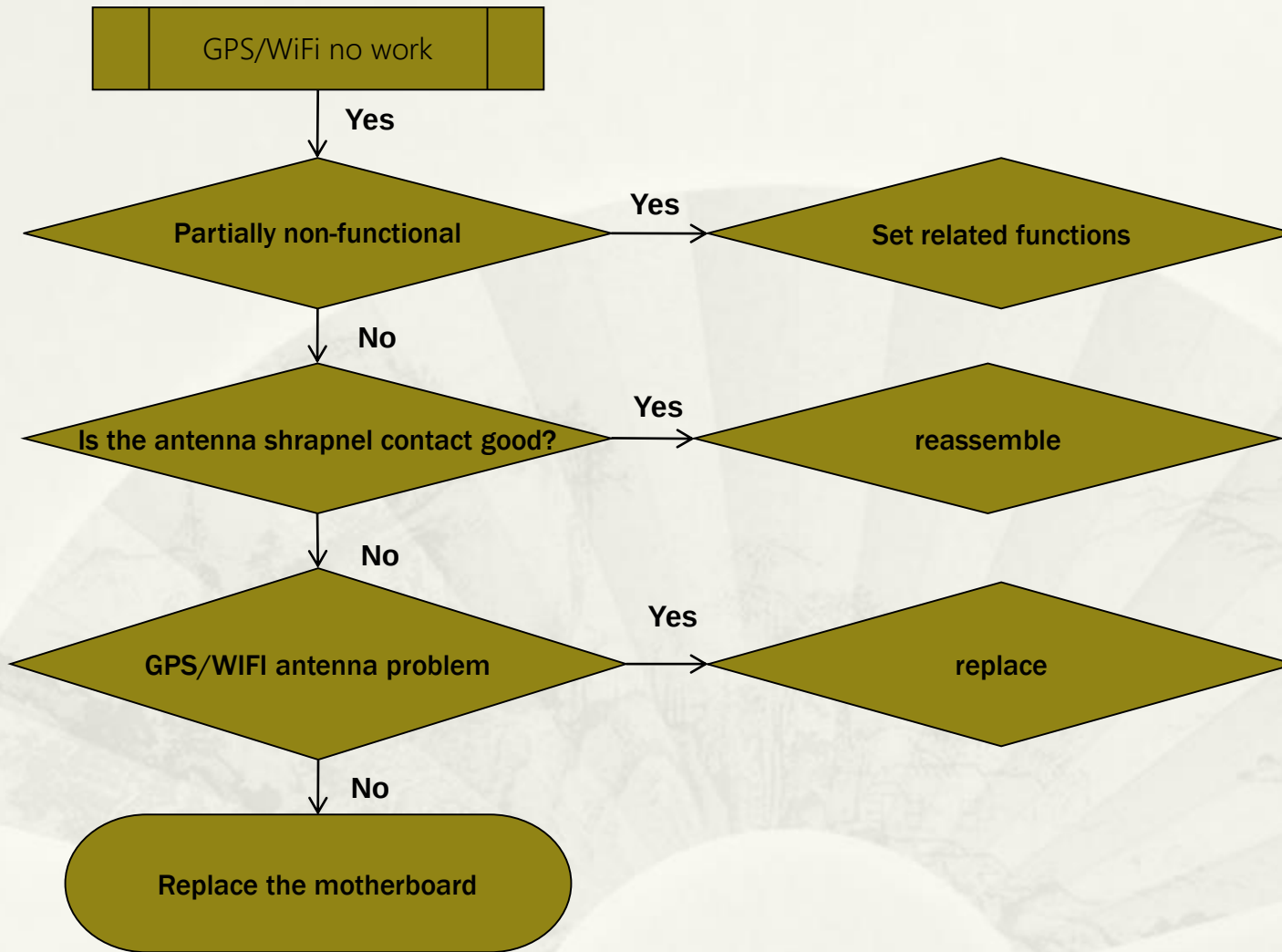
12. Distance sensing failure analysis



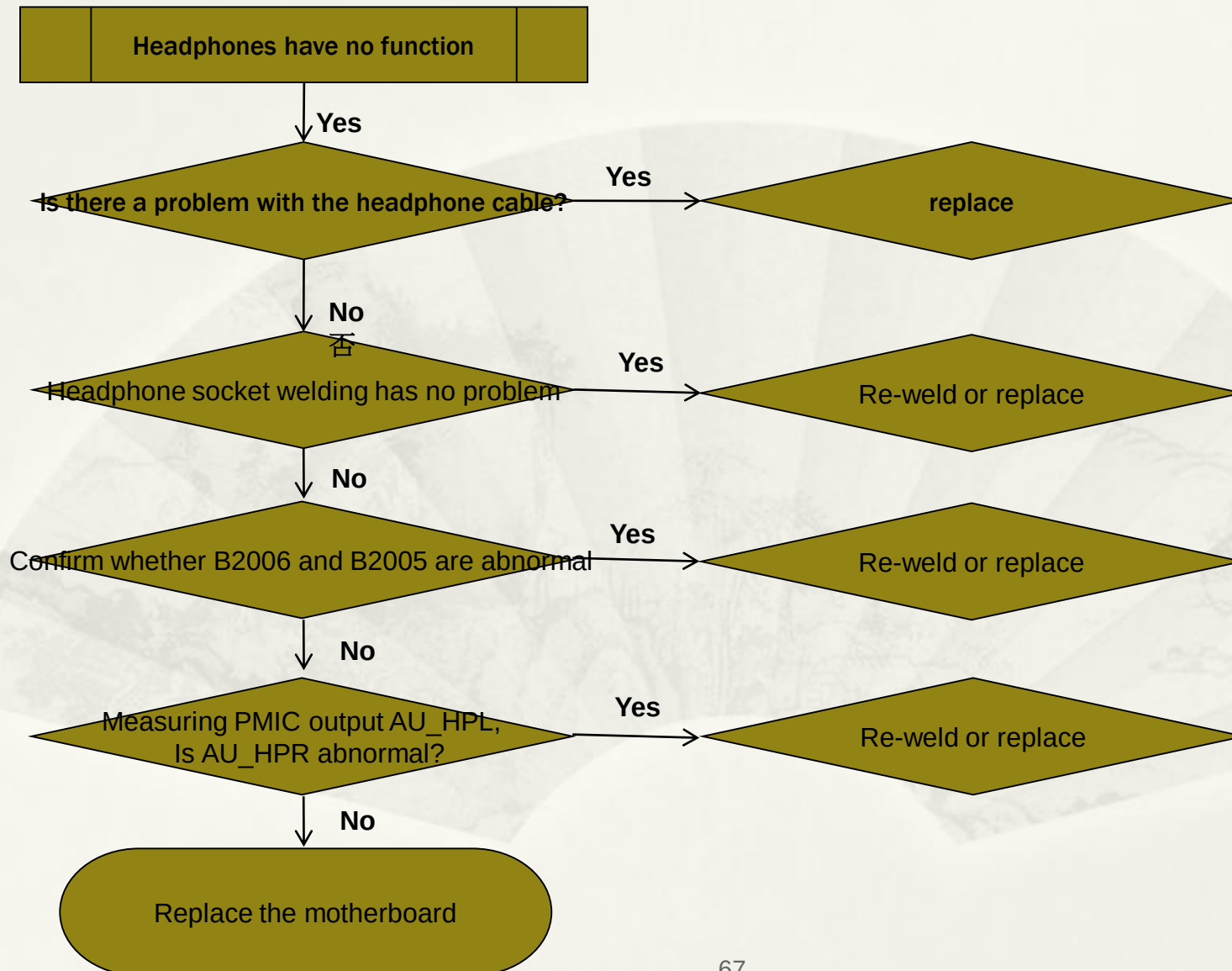
13. No signal failure analysis



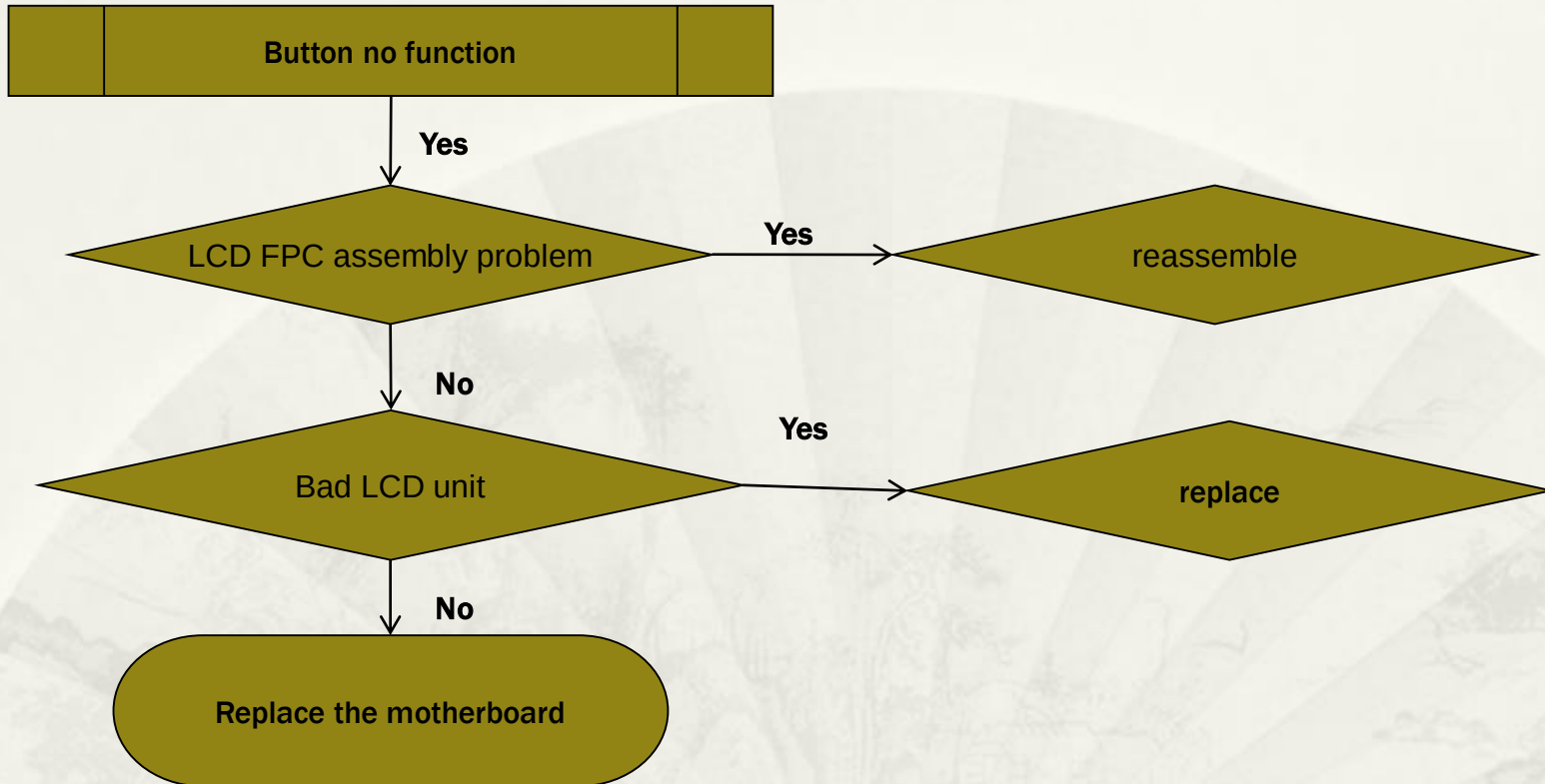
14. GPS/WiFi Failure analysis



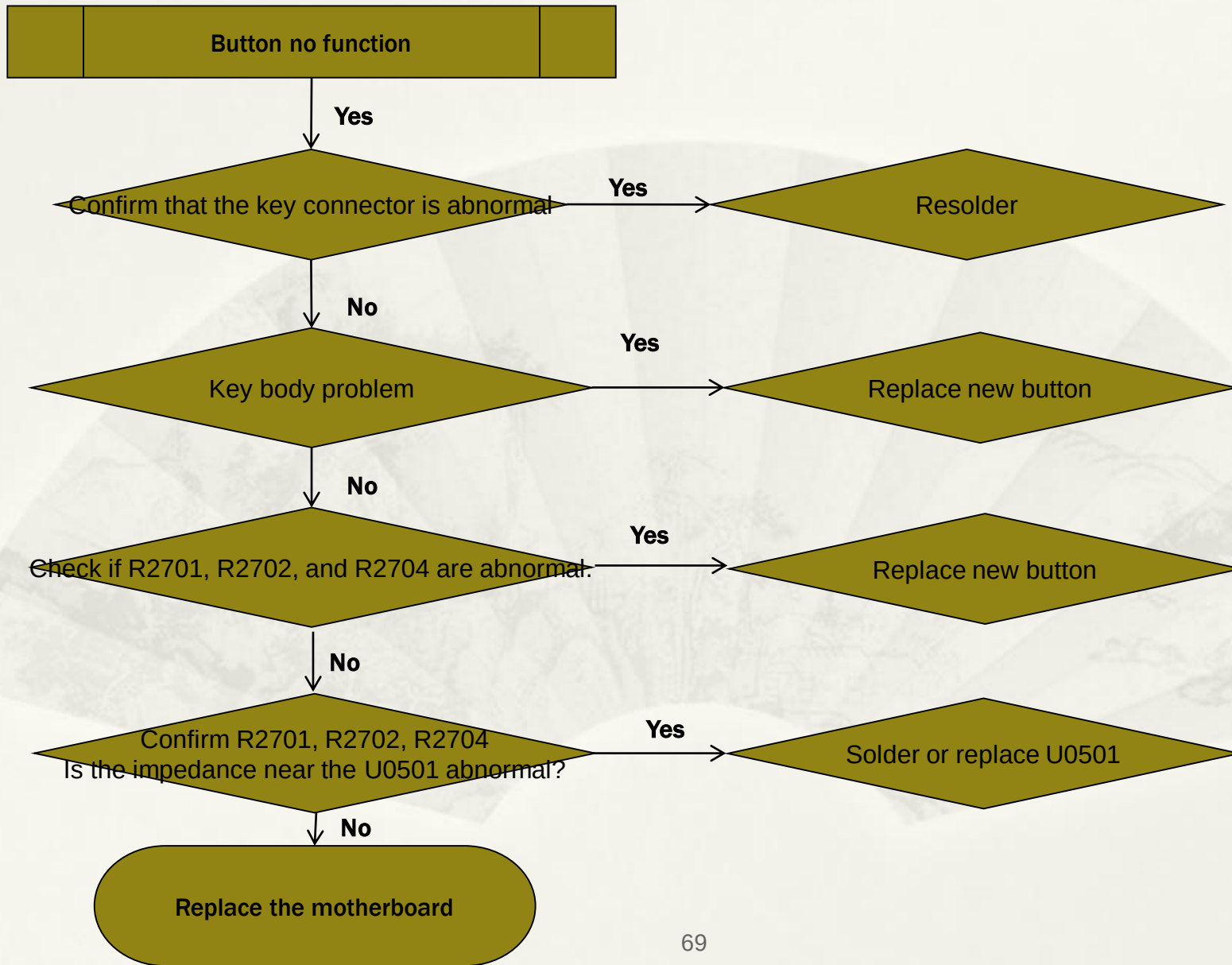
15. Headphone problem analysis



16. Virtual button or TP failure analysis



17. Key failure analysis





Thank you !